



**California Regional Water Quality Control Board
Central Coast Region**



Linda S. Adams.
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Environmental Protection

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Arnold Schwarzenegger
Governor

April 6, 2010

Jay Baksa, City Administrator
South County Regional Wastewater Authority
7351 Rosanna Street
Gilroy, CA 95020

Dear Mr. Baksa:

**WASTE DISCHARGE REQUIREMENTS, SOUTH COUNTY REGIONAL WASTEWATER
AUTHORITY WASTEWATER TREATMENT AND RECLAMATION FACILITY, SANTA CLARA
COUNTY (NPDES PERMIT NO. CA0049964) - ORDER NO. R3-2010-0009**

At its public meeting on March 18, 2010, the Central Coast Water Board adopted Order No. R3-2010-0009, Waste Discharge Requirements for the South County Regional Wastewater Authority (reissued NPDES Permit No. CA0049964).

The permit will also be posted online at:

http://www.waterboards.ca.gov/centralcoast/board_decisions/adopted_orders/index.shtml

If you have any questions, please call **Mike Higgins at 805/542-4649** or Burton Chadwick at 805/542-4786.

Sincerely,


Roger W. Briggs
Executive Officer

Attachments: WDR Order No. R3-2010-0009

cc: (with electronic attachments)

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SWRCB – DWQ (NPDES_Wastewater@waterboards.ca.gov and dmr@waterboards.ca.gov)
Jae Kim, Tetra Tech (jae.kim@tetratech-ffx.com)

cc: (without attachment)

SCRWA Interested parties list

S:\NPDES\NPDES Facilities\Santa Clara Co\SCRWA\Order No. R3-2010-0009\ADOPTED COVER LETTER.doc

California Environmental Protection Agency



California Regional Water Quality Control Board Central Coast Region



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DRAFT ORDER NO. R3-2010-0009
NPDES NO. CA0049964
WDID No. 3 430100001

WASTE DISCHARGE REQUIREMENTS FOR THE SOUTH COUNTY REGIONAL WASTEWATER AUTHORITY SOUTH COUNTY REGIONAL WASTEWATER TREATMENT AND RECLAMATION FACILITY

The following Discharger is subject to waste discharge requirements as set forth in this Order.

Table 1. Discharger Information

Discharger	South County Regional Wastewater Authority
Name of Facility	South County Wastewater Treatment and Reclamation Facility
Facility Address	1500 Southside Drive
	Gilroy, CA 95020
	Santa Clara County
The U.S. Environmental Protection Agency (USEPA) and the Regional Water Quality Control Board have classified this discharge as a major discharge.	

Discharges by the **South County Regional Wastewater Authority** from the discharge points identified below are subject to waste discharge requirements as set forth in this Order:

Table 2. Discharge Location

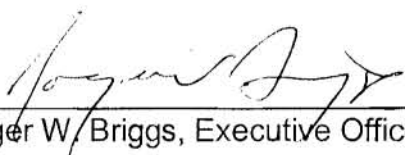
Discharge Point	Effluent Description	Discharge Point Latitude	Discharge Point Longitude	Receiving Water
001	Secondary Treated Municipal Wastewater	36° 58' 50" N	121° 32' 00" W	Land Application to 37 Percolation Ponds (Adjacent to Llagas Creek)
002	Tertiary Treated Municipal Wastewater	36° 56' 52" N	121° 30' 43" W	Pájaro River
003	Tertiary Treated Municipal Wastewater	---	---	Title 22 Reclamation Use

Table 3. Administrative Information

This Order was adopted by the Regional Water Quality Control Board on:	March 18, 2010
This Order shall become effective on:	March 18, 2010
This Order shall expire on:	March 18, 2015
The Discharger shall file a Report of Waste Discharge in accordance with Title 23, California Code of Regulations (CCR), as application for issuance of new waste discharge requirements no later than:	180 days prior to the Order expiration date

THEREFORE, IT IS HEREBY ORDERED, that this Order supersedes Order No. R3-2004-0099 except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the federal CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order.

I, Roger Briggs Executive Officer, do hereby certify that this Order, with all attachments, is a full, true, and correct copy of an Order adopted by the Regional Water Board on March 18, 2010.



Roger W. Briggs, Executive Officer

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I. FACILITY INFORMATION

The following Discharger is subject to waste discharge requirements as set forth in this Order:

Table 4. Facility Information

Discharger	South County Regional Wastewater Authority
Name of Facility	South County Regional Wastewater Treatment and Reclamation Facility
Facility Address	1500 Southside Drive
	Gilroy, California 93446
	Santa Clara County
Facility Contact, Title, and Phone	Saeid Vaziry, Chief Engineer (408) 846-8842
Mailing Address	7351 Rosanna Street, Gilroy, California 95020
Type of Facility	Publicly Owned Treatment Works (POTW)
Facility Design Flow	8.5 million gallons per day (MGD) (average dry weather, secondary treatment capacity)
	9.0 MGD (tertiary treatment capacity)

II. FINDINGS

The Regional Water Board finds:

A. Background. The South County Regional Wastewater Authority (SCRWA, hereinafter the Discharger) is discharging treated municipal wastewater pursuant to Order No. R3-2004-0099 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0049964. The Discharger submitted a Report of Waste Discharge, dated March 9, 2009, and applied to renew its NPDES permit to discharge up to 8.5 MGD (average dry weather flow) of secondary treated wastewater from the South County Regional Wastewater Treatment and Reclamation Facility (Reclamation Plant). The application was deemed complete on May 14, 2009, by Regional Water Board staff.

For the purposes of this Order, references to the "discharger" or "permittee" in applicable federal and State laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

B. Facility Description. The Cities of Gilroy and Morgan Hill own and maintain wastewater collection systems within each respective city. An interceptor sewer owned and maintained by the City of Gilroy conveys wastewater from the two cities to the Reclamation Plant, which is owned by the SCRWA and operated by CH2M Hill OMI. The reclamation plant's design secondary treatment capacities are 8.5 million gallons per day (MGD, average dry weather flow) and 10.2 MGD (average wet weather flow) and its tertiary treatment capacity is 9.0 MGD. Secondary treated wastewater is land applied at Discharge Point 001 to percolation ponds adjacent to Llagas Creek. Tertiary treated wastewater is used to irrigate farmlands and may be discharged at Discharge Point 002 to the Pájaro River.

Secondary treatment is accomplished with two parallel treatment sequences, each including four pre-anoxic basins, an aerated oxidation ditch, and a secondary clarifier. A post-anoxic basin and a reaeration basin serve both sequences before disposal in the ponds or delivery to tertiary treatment, which includes filtration and chlorination/dechlorination processes.

Secondary treated wastewater is distributed from Discharge Point 001 to 37 percolation ponds, which are adjacent to Llagas Creek. The Discharger irrigates onsite landscaping with reclaimed wastewater, which also supplies the fire protection system. The Discharger delivers most of the reclaimed water offsite for irrigation or industrial cooling by the City of Gilroy Parks, the Calpine Cogeneration Facility, the Gilroy Golf Course, the Eagle Ridge Golf Course, Obata Farms, and the McCarthy Business Park. Under emergency conditions during wet weather events, tertiary treated wastewater may also be discharged at Discharge Point 002 to the Pájaro River. During the 2004 – 2008 time period, there were no discharges at this outfall. During the five year term of this Order, ultraviolet disinfection capability and a new outfall line will be installed to manage discharges to the Pájaro River.

Attachment B provides a map of the area around the Reclamation Plant and disposal ponds. Attachment C provides a flow schematic of the facility.

- C. Legal Authorities.** This Order is issued pursuant to CWA section 402 and implementing regulations adopted by the USEPA and Chapter 5.5, Division 7 of the California Water Code (CWC), commencing with section 13370. It shall serve as an NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the CWC, commencing with section 13260.
- D. Background and Rationale for Requirements.** The Regional Water Board developed the requirements of this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information, including a site visit on April 23, 2009. The Fact Sheet (Attachment F), which contains background information and rationale for the Order's waste discharge requirements, is hereby incorporated into this Order and constitutes part of the Findings for this Order. Attachments A through E are also incorporated into this Order.
- E. California Environmental Quality Act (CEQA).** Pursuant to Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of the CEQA, Public Resources Code sections 21100-21177.
- F. Technology-Based Effluent Limitations.** CWA Section 301 (b) and USEPA's NPDES regulations at 40 CFR 122.44 require that permits include, at a minimum, conditions meeting applicable technology-based requirements and any more stringent effluent limitations necessary to meet applicable water quality standards. Discharges authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards established at 40 CFR Part 133 and Best Professional Judgment (BPJ) in accordance with 40 CFR 125.3. The Fact Sheet (Attachment F) provides the basis and rationale for development of technology-based effluent limitations.

G. Water Quality-Based Effluent Limitations. CWA Section 301 (b) and NPDES regulations at 40 CFR 122.44 (d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards.

NPDES regulations at 40 CFR 122.44 (d)(1)(i) mandate that permits include effluent limitations for all pollutants that are or may be discharged at levels with the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential is established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) USEPA criteria guidance under CWA section 304 (a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed State criterion or policy interpreting the State's narrative criterion, supplemented with other relevant information, as provided at 40 CFR 122.44 (d)(1)(vi).

H. Water Quality Control Plans. The Regional Water Board has adopted a *Water Quality Control Plan for the Central Coast Region* (the Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for receiving waters within the Region. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which establishes State policy that all waters, with certain exceptions, are suitable or potentially suitable municipal or domestic drinking water supplies. Beneficial uses established by the Basin Plan for the Pajaro River are presented in Table 5, below.

Table 5. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
002	Pájaro River	• Municipal and domestic water supply (MUN) • Agricultural supply (AGR) • Industrial process supply (PRO) • Groundwater recharge (GWR) • Contact (REC-1) and Non-contact (REC-2) water recreation • Wildlife habitat (WILD) • Cold freshwater habitat (COLD) • Warm freshwater habitat (WARM) • Migration of aquatic organisms (MIGR) • Spawning, reproduction, and/or early development (SPWN) • Freshwater replenishment (FRESH) • Commercial and sport fishing (COMM).

Beneficial uses established by the Basin Plan for groundwaters include municipal and domestic water supply, agricultural supply, and industrial process supply. This Order's waste discharge requirements implement the Basin Plan. The Order should thereby ensure regulated waste discharges do not impair or threaten to impair the beneficial uses of the Pájaro River, Llagas Creek, or groundwater.

- I. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, USEPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the State. The CTR was amended on February 13, 2001. These rules contain water quality criteria for priority pollutants that are applicable to discharges from the Reclamation Plant
- J. **State Implementation Policy.** On March 2, 2000, the State Water Board adopted the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (State Implementation Policy or SIP). The SIP became effective on April 28, 2000 with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Regional Water Board in the Basin Plan. The SIP became effective on May 18, 2000 with respect to the priority pollutant criteria promulgated by the USEPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.
- K. **Compliance Schedules and Interim Requirements.** Section 2.1 of the SIP provides that, based on a Discharger's request and demonstration that it is infeasible for an existing Discharger to achieve immediate compliance with an effluent limitation derived from a CTR criterion, compliance schedules may be allowed in an NPDES permit. Unless an exception has been granted under section 5.3 of the SIP, a compliance schedule may not exceed 5 years from the date that the permit is issued or reissued, nor may it extend beyond 10 years from the effective date of the SIP (or May 18, 2010) to establish and comply with CTR criterion-based effluent limitations. Where a compliance schedule for a final effluent limitation exceeds one year, the Order must include interim numeric limitations for that constituent or parameter. Where allowed by the Basin Plan, compliance schedules and interim effluent limitations or discharge specifications may also be granted to allow time to implement a new or revised water quality objective. This Order does not include compliance schedules or interim effluent limitations.
- L. **Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised State and tribal water quality standards (WQS) become effective for CWA purposes [65 Fed. Reg. 24641 (April 27, 2000) (codified at 40 CFR 131.21)]. Under the revised regulation (also known as the Alaska Rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.
- M. **Stringency of Requirements for Individual Pollutants.** This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based effluent limitations consist of restrictions for biochemical oxygen demand (BOD₅), total suspended solids (TSS), and pH, and are discussed in section IV.B

of the Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements.

WQBELs have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant WQBELs were derived from the CTR, the CTR is the applicable standard pursuant to NPDES regulations at 40 CFR 131.38. The scientific procedures for calculating the individual WQBELs for priority pollutants are based on the CTR and the SIP, which was approved by USEPA on May 18, 2000. All beneficial uses and water quality objectives contained in the Basin Plan were approved under State law and submitted to and approved by USEPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to NPDES regulations at 40 CFR 131.21 (c) (1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

- N. Antidegradation Policy.** NPDES regulations at 40 CFR 131.12 require that State water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16, which incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that the existing quality of waters be maintained unless degradation is justified based on specific findings. The Basin Plan implements and incorporates by reference both the State and federal antidegradation policies. As discussed in Section III.C.5 of the Fact Sheet, the permitted discharge is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution No. 68-16.
- O. Anti-Backsliding Requirements.** CWA sections 402 (o) (2) and 303 (d) (4) and NPDES regulations at 40 CFR 122.44 (l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. All effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order.
- P. Endangered Species Act.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code sections 2050 to 2097) or the federal Endangered Species Act (16 U.S.C.A. sections 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the State. The Discharger is responsible for meeting all requirements of State and federal law regarding threatened and endangered species.
- Q. Monitoring and Reporting.** NPDES regulations at 40 CFR 122.48 require that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 also authorize the Regional Water Board to

require technical and monitoring reports. The Monitoring and Reporting Program, provided as Attachment E to the Order, establishes monitoring and reporting requirements to implement federal and State requirements.

R. Standard and Special Provisions. Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR 122.42, are provided in Attachment D. The Discharger must comply with all standard provisions and with those additional conditions that are applicable pursuant to 40 CFR 122.42. The Regional Water Board has also included in this Order special provisions applicable to the Discharger. A rationale for the special provisions contained in this Order is provided in the attached Fact Sheet.

S. Recycled Water Policy. A priority of the Strategic Plan Update 2008-2012 for the Water Boards is to increase sustainable local water supplies available for existing and future beneficial uses by 1,725,000 acre-feet per year, in excess of 2002 levels, by 2015, and to ensure adequate water flows for fish and wildlife habitat. The State Water Resources Control Board (State Water Board) adopted the Recycled Water Policy (Resolution No. 2009-0011) on February 3, 2009. The Recycled Water Policy is intended to support the Strategic Plan priority. Increasing public acceptance and promoting the use of recycled water is a means towards achieving sustainable local water supplies and can result in reduction in greenhouse gases, a significant driver of climate change.

The Recycled Water Policy calls for the development of regional groundwater basin/sub-basin salt/nutrient management plans. The State Water Board recognizes that local water and wastewater entities, together with other local salt/nutrient contributors to the State's groundwaters, will fund and develop salt and nutrient management plans for each basin/sub-basin in California. Plan development will be locally driven and controlled, collaborative, and will be open to all stakeholders, including Regional Water Board staff. Plans will comply with CEQA. State Water Board's recognition of local control is in response to the December 19, 2008 letter from statewide water and wastewater entities, attached to Resolution No. 2009-0011, which adopted the Policy.

It is the intent of the Recycled Water Policy that salts and nutrients from all sources be managed on a basin-wide or watershed-wide basis in a manner that ensures attainment of water quality objectives and protection of beneficial uses. The State Water Board finds that the appropriate way to address salt and nutrient issues is through the development of regional or sub-regional salt and nutrient management plans rather than through imposing requirements solely on individual projects. The Central Coast Water Board finds that a combination of regional management plans and individual or programmatic project requirements may be necessary to protect beneficial uses.

One of the primary components of the required regional salt/nutrient management plans is the development and implementation of groundwater basin/sub-basin monitoring programs. As specified in the Recycled Water Policy, salt/nutrient contributing

stakeholders will be responsible for conducting, compiling, and reporting the monitoring data once the regional groundwater monitoring programs are developed.

Technical reports and data in Central Coast Water Board files document widespread and increasing salt and nutrient pollution in groundwater basins throughout the Central Coast Region, including the Pájaro River groundwater basin and sub-basins.

- T. Provisions and Requirements Implementing State Law.** The provisions and requirements in subsections IV.B, IV.C, and V.B of this Order are included to implement State law only. These provisions and requirements are not required or authorized under the federal CWA; consequently, violations of these provisions and requirements are not subject to the enforcement remedies that are available for NPDES violations.
- U. Notification of Interested Parties.** The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of notification are provided in the Fact Sheet of this Order.
- V. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the public hearing are provided in the Fact Sheet of this Order.

III. DISCHARGE PROHIBITIONS

- A.** Discharge of secondary treated wastewater at a location or in a manner other than as described by this Order at Discharge Point 001 and of tertiary treated wastewater as described by this Order at Discharge Points 002 and 003 is prohibited.
- B.** The discharge of any waste not specifically regulated by this Order is prohibited.
- C.** Creation of a condition of pollution, contamination, or nuisance, as defined by Section 13050 of the CWC Water Code, is prohibited.
- D.** The overflow or bypass of wastewater from the Discharger's collection, treatment, or disposal facilities and the subsequent discharge of untreated or partially treated wastewater, except as provided for in Attachment D, Standard Provision I. G (Bypass), is prohibited.
- E.** Discharges of sludge, residues, or any other wastes into surface waters or into any area where they may enter surface water, are prohibited.
- F.** Average dry weather influent flow from the wastewater treatment facility (the average daily flow in the three driest months of each year) shall not exceed the facility's dry weather treatment capacity of 8.5 MGD. Average wet weather influent flow from the wastewater treatment facility (the average daily flow in the three wettest months of each year) shall not exceed the facility's wet weather treatment capacity of 10.8 MGD.

- G. The discharge of fecal coliform bacteria originating from human sources at Discharge Point 002 to the Pajaro River is prohibited.

IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Effluent Limitations – Discharge Point 002

1. Final Effluent Limitations

- a. **Conventional and Non-Conventional Pollutants.** The Discharger shall maintain compliance with the following effluent limitations at Discharge Point 002 (discharge to the Pajaro River), with compliance measured at Monitoring Location EFF-002, as described in the attached Monitoring and Reporting Program (MRP).

Table 6. Effluent Limitations for Conventional and Non-Conventional Pollutants

Constituent	Units	Effluent Limits			
		Average Monthly	Average Weekly	Maximum Daily	12-Month Moving Avg
Flow	MGD	---	---	9.0	---
BOD ₅	mg/L	10	---	20	---
TSS	mg/L	10	---	20	---
pH	s.u.	6.5 – 8.3 at all times			
Nitrate	mg/L N	5	---	10	---
Unionized Ammonia	mg/L N	0.025	---	0.050	---
TDS	mg/L	1,000	---	---	---
Chloride	mg/L	250	---	---	---
Sodium	mg/L	200	---	---	---
Sulfate	mg/L	250	---	---	---
Boron	mg/L	1.0	---	---	---
Turbidity	NTU	- Daily average turbidity shall be less than or equal to 2 NTU, - Turbidity shall be less than 10 NTU at all times, and - Turbidity shall not exceed 5 NTU for more than 5 percent of the time.			
Coliform Bacteria	Organisms/100 mL	- The 7-day median concentration shall be less than 2.2 organisms/100 mL, - Coliform concentrations shall not exceed 23 organisms/100 mL in more than one sample in any thirty day period, and - Coliform concentrations shall be less than 240 organisms/100 mL at all times.			
Chlorine	mg/L	If chlorine is used for disinfection, a CT value (chlorine concentration times modal contact time) of not less than 450 mg-min/L shall be maintained at all times with a modal contact time of at least 90 minutes based on a discharge rate of 9.0 MGD. Chlorine concentrations in effluent shall be non-detectable at all times as determined by amperometric titration or another			

Constituent	Units	Effluent Limits			
		Average Monthly	Average Weekly	Maximum Daily	12-Month Moving Avg
		equally sensitive method.			

- b. The average monthly percent removal of BOD₅ and TSS shall not be less than 85 percent.
- c. **Toxic Pollutants.** The Discharger shall maintain compliance with the following effluent limitations for toxic pollutants at Discharge Point 002, with compliance measured at Monitoring Location EFF-002, as described in the attached MRP.

Table 7. Effluent Limitations for Toxic Pollutants – Discharge Point 002

Constituent	Units	Effluent Limits	
		Average Monthly	Maximum Daily
Lead	µg/L	2.1	4.2
Copper	µg/L	20	42
Chlorodibromomethane	µg/L	0.40	0.80

2. Additional Specifications for Discharges of Tertiary Treated Wastewater to the Pajaro River at Discharge Point 002

- a. Discharge of tertiary treated wastewater to the Pajaro River shall occur only during the months of November through April, on an as needed basis, to facilitate the proper maintenance and safe operation of the percolation ponds.
- b. Discharges to the Pajaro River shall occur only when flow in the Pajaro River is greater than 180 MGD, as measured at a gauging station near the point of discharge, and when flow in the Pajaro River is below 6,004 MGD, as measured at the Chittenden gauging station.

2. Interim Effluent Limitations

This section of the standardized permit template is not applicable.

B. Land Discharge Specifications

1. The Discharger shall comply with the following specifications at Discharge Point 001(land application to the percolation ponds), with compliance measured at Monitoring Location EFF-001, as described in the attached Monitoring and Reporting Program (MRP).

Table 8. Land Discharge Specifications – Discharge Point 001

Constituent	Units	Effluent Limits			
		Average Monthly	Average Weekly	Maximum Daily	12-Month Moving Avg

Discharge Point 001					
BOD ₅	mg/L	30	45	---	
TSS	mg/L	30	45	---	
Nitrate	mg/L N	5	---	10	
TDS	mg/L	---	---	---	900
Chloride	mg/L	---	---	---	200
Sodium	mg/L	---	---	---	175
Sulfate	mg/L	---	---	---	150
Boron	mg/L	---	---	---	1.0
pH	s.u.	6.5 – 8.3 at all times			

- a. The average monthly percent removal of BOD₅ and TSS shall not be less than 85 percent.
2. Additional Specifications for Discharges of Secondary Treated Wastewater to the Percolation Ponds at Discharge Point 001
 - a. Freeboard shall always exceed 2 feet in all percolation ponds.
 - b. Extraneous surface drainage shall be excluded from all percolation ponds.
 - c. Irrigation beds and designated percolation ponds shall be disked or plowed at least annually.
 - d. Wastewater shall be confined to land owned or controlled by the Discharger.
 - e. Wastewater shall be confined within bermed areas.
 - f. Wastewater application rates to the percolation ponds shall be consistent with accepted engineering practice.
 - g. Percolation ponds shall be dried to field moisture conditions between applications.
 - h. A pathway shall be maintained along the dike between the designated percolation areas and Llagas Creek to allow inspections.
 - i. The wastewater treatment facility, including the percolation ponds, shall be managed to minimize mosquito breeding habitat.

C. Reclamation Specifications

Reclamation use of tertiary treated wastewater shall adhere to applicable requirements of CWC sections 13500 – 13577 (Water Reclamation) and of California Code of Regulations (CCR) Title 22, sections 60301 – 60357 (Water Recycling Criteria).

V. RECEIVING WATER LIMITATIONS

A. Surface Water Limitations

Receiving water limitations are based on water quality objectives contained in the Basin Plan and are a required part of this Order. Discharges from the wastewater treatment facility shall not cause the following conditions in Llagas Creek and the Pajaro River.

1. Waters shall be free of coloration that causes nuisance or adversely affects beneficial uses. Coloration attributable to materials of waste origin shall not be greater than 15 units or 10 percent above natural background color, whichever is greater.
2. Waters shall not contain taste or odor-producing substances in concentrations that impart undesirable tastes or odors to fish flesh or other edible products of aquatic origin, that cause nuisance, or that adversely affect beneficial uses.
3. Waters shall not contain floating material, including solids, liquids, foams, and scum, in concentrations that cause nuisance or adversely affect beneficial uses.
4. Waters shall not contain suspended material in concentrations that causes nuisance or adversely affects beneficial uses.
5. Waters shall not contain settleable material in concentrations that result in deposition of material that causes nuisance or adversely affects beneficial uses.
6. Waters shall not contain oils, greases, waxes, or other similar materials in concentrations that result in a visible film or coating on the surface of the water or on objects in the water, that cause nuisance, or that otherwise adversely affect beneficial uses.
7. Waters shall not contain biostimulatory substances in concentrations that promote aquatic growths to the extent that such growths cause nuisance or adversely affect beneficial uses.
8. The suspended sediment load and suspended sediment discharge rate to surface waters shall not be altered in such a manner as to cause nuisance or adversely affect beneficial uses.
9. Waters shall be free of changes in turbidity that cause nuisance or adversely affect beneficial uses. Increase in turbidity attributable to controllable water quality factors shall not exceed the following limits.
 - a. Where natural turbidity is between 0 and 50 Jackson Turbidity Units (JTU), increases shall not exceed 20 percent.
 - b. Where natural turbidity is between 50 and 100 JTU, increases shall not exceed 10 percent.

- c. Where natural turbidity is greater than 100 JTU, increases shall not exceed 10 percent.
10. To protect cold freshwater habitat, the pH value shall not be depressed below 7.0 nor raised above 8.3. The change in normal ambient pH levels shall not exceed 0.5 in fresh water.
11. To protect cold freshwater habitat, dissolved oxygen concentrations in receiving waters shall not be reduced below 7.0 mg/L at any time. If background concentration of dissolved oxygen in receiving waters is less than 7.0 mg/L, then discharges shall not reduce dissolved oxygen concentrations.
12. Natural temperature of receiving waters shall not be altered unless it can be demonstrated to the satisfaction of the Regional Water Board that such alteration in temperature does not adversely affect beneficial uses. To protect cold and warm freshwater habitat, at no time or place shall the temperature of any water be increased by more than 5° F above the natural receiving water temperature.
13. All waters shall be maintained free of toxic substances in concentrations which are toxic to, or which produce detrimental physiological responses in, human, plant, animal, or aquatic life. Survival of aquatic life in surface waters subjected to a waste discharge or other controllable water quality conditions shall not be less than that for the same water body in areas unaffected by the waste discharge.
14. The discharge of wastes shall not cause concentrations of unionized ammonia (NH₃) to exceed 0.025 mg/L (as N) in the receiving water.
15. No individual pesticide or combination of pesticides shall reach concentrations that adversely affect the beneficial uses of the receiving water. There shall be no increase in pesticide concentrations found in bottom sediments or aquatic life. For waters where existing concentrations are presently nondetectable or where beneficial uses would be impaired by concentrations in excess of nondetectable levels, total identifiable chlorinated hydrocarbon pesticides shall not be present at concentrations detectable within the accuracy of analytical methods as prescribed in *Standard Methods for the Examination of Water and Wastewater*, latest edition, or other equivalent methods approved by the Executive Officer.
16. Waters shall not contain organic substances in concentrations greater than the following.
- | | |
|-------------------------------------|------------|
| Methylene Blue Activated Substances | 0.2 mg/L |
| Phenols | 0.1 mg/L |
| PCBs | 0.3 µg/L |
| Phthalate Esters | 0.002 µg/L |
17. Radionuclides shall not be present in concentrations that are deleterious to human, plant, animal, or aquatic life; or result in the accumulation of radionuclides in the food web to an extent which presents a hazard to human, plant, animal, or aquatic life. To protect the municipal and domestic supply beneficial use, receiving waters shall

not contain concentrations of radionuclides greater than the maximum contaminant levels (MCLs) for radioactivity presented in Table 4 of Title 22 CCR Division 4, Chapter 15, Article 5.

18. To protect the municipal and domestic supply beneficial use, receiving waters shall not contain concentrations of chemical constituents in excess of the primary maximum contaminant levels (MCLs) specified for drinking water in Table 64431-A (Primary MCLs for Inorganic Chemicals) and Table 64444-A (Primary MCLs for Organic Chemicals) of Title 22 CCR, Division 4, Chapter 15.
19. To protect the water contact recreation beneficial use, fecal coliform concentration, based on a minimum of not less than five samples for any 30-day period, shall not exceed a log mean of 200 per 100 mL, nor shall more than 10 percent of samples collected during any 30-day period exceed 400 per 100 mL.
20. Receiving waters shall not contain concentrations of chemical constituents in amounts that adversely affect the agricultural beneficial use. Interpretation of "adverse effect" shall be based on the University of California Agricultural Extension Service guidelines presented in Table 3-3 of the Basin Plan.

Waters used for irrigation and livestock watering shall not contain pollutants in excess of the "Water Quality for Agricultural Water Use" established by Table 3-4 of the Basin Plan.

21. To protect cold and warm freshwater habitat beneficial uses, receiving waters shall not contain metals in excess of the following concentrations, established by Table 3-5 of the Basin Plan.

Metal	Receiving Water Hardness > 100 mg/L CaCO ₃	Receiving Water Hardness < 100 mg/L CaCO ₃
Cadmium ^[1]	0.03 mg/L	0.004 mg/L
Chromium	0.05 mg/L	0.05 mg/L
Copper	0.03 mg/L	0.01 mg/L
Lead	0.03 mg/L	0.03 mg/L
Mercury ^[2]	0.0002 mg/L	0.0002 mg/L
Nickel ^[3]	0.4 mg/L	0.1 mg/L
Zinc	0.2 mg/L	0.004 mg/L

^[1] Lower cadmium values not to be exceeded for crustaceans and waters designated SPWN are 0.003 mg/L in hard water and 0.0004 mg/L in soft water.

^[2] Total mercury values should not exceed 0.05 mg/L as an average value; maximum acceptable concentration of total mercury in any aquatic organism is a total B.O.D. burden of 0.5 mg/L wet weight.

^[3] Value cited as objective pertains to nickel salts (not pure metallic nickel).

22. The following surface water quality objectives for the Pajaro River at Chittenden, established by Table 3-7 of the Basin Plan, shall not be exceeded.

	TDS	Chloride	Sulfate	Boron	Sodium
Pajaro River	1000 mg/L	250 mg/L	250 mg/L	1.0 mg/L	200 mg/L

These objectives are annual mean values and are objectives based on preservation of existing quality or water quality enhancement believed attainable following control of point sources

B. Groundwater Limitations

Activities at and discharges from the treatment facility shall not cause exceedance/deviation from the following water quality objectives for groundwater established by the Basin Plan.

1. Groundwater shall not contain taste or odor producing substances in concentrations that adversely affect beneficial uses.
2. Radionuclides shall not be present in concentrations that are deleterious to human, plant, animal, or aquatic life; or result in the accumulation of radionuclides in the food web to an extent that presents a hazard to human, plant, animal, or aquatic life. To protect the municipal and domestic supply beneficial use, in no circumstance shall receiving waters contain concentrations of radionuclides in excess of the maximum contaminant levels (MCLs) for radioactivity presented in Table 4 of Title 22, CCR, Division 4, Chapter 15, Article 5.
3. To protect the municipal and domestic supply beneficial use, the median concentration of coliform organisms in groundwater, over any seven-day period, shall be less than 2.2 organisms per 100 milliliters.
4. To protect the municipal and domestic supply beneficial use, groundwater shall not contain concentrations of chemical constituents in excess of the primary maximum contaminant levels (MCLs) specified for drinking water in Table 64431-A (Primary MCLs for Inorganic Chemicals) and Table 64444-A (Primary MCLs for Organic Chemicals) of Title 22, CCR, Division 4, Chapter 15. Note that the MCL for nitrate is 10 mg/L expressed as N.
5. Groundwater shall not contain concentrations of chemical constituents in amounts that adversely affect the agricultural beneficial use. Interpretation of "adverse effect" shall be based on the University of California Agricultural Extension Service guidelines presented in Table 3-3 of the Basin Plan. Waters used for irrigation and livestock watering shall not contain pollutants in excess of the "Water Quality for Agricultural Water Use" established by Table 3-4 of the Basin Plan.

VI. PROVISIONS

A. Standard Provisions

The Discharger shall comply with all Standard Provisions included as Attachment D of this Order.

B. Monitoring and Reporting Program (MRP) Requirements

The Discharger shall comply with the Monitoring and Reporting Program, and future revisions thereto, in Attachment E of this Order. All monitoring shall be conducted according to 40 CFR Part 136, *Guidelines Establishing Test Procedures for Analysis of Pollutants*.

C. Special Provisions

1. Reopener Provisions

- a. This permit may be reopened and modified in accordance with NPDES regulations at 40 CFR 122 and 124, as necessary, to include additional conditions or limitations based on newly available information or to implement any USEPA approved, new, State water quality objective.

2. Special Studies, Technical Reports and Additional Monitoring Requirements

a. Toxicity Reduction Requirements

As indicated in Section V.D. of the MRP, accelerated monitoring for toxicity is required upon the detection of acute toxicity or the chronic toxicity trigger value of 1 TUC is exceeded. The Discharger shall conduct a Toxicity Reduction Evaluation (TRE) in accordance with the Discharger's TRE Workplan upon consistent detection of toxicity in the effluent during accelerated testing.

A TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A TOXICITY IDENTIFICATION EVALUATION (TIE) may be required as part of the TRE, if appropriate. A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases - characterization, identification, and confirmation using aquatic organism toxicity tests. The TRE shall include all reasonable steps to identify the source of toxicity. The Discharger shall take all reasonable steps to reduce toxicity to the required level once the source of toxicity is identified.

The Discharger shall maintain a Toxicity Reduction Evaluation (TRE) Workplan, which describes steps that the Discharger intends to follow in the event that a toxicity effluent limitation established by this Order is exceeded in the discharge. The workplan shall be prepared in accordance with current technical guidance and reference material, including EPA/600/2-88-070 (for industrial discharges) or EPA/600/2-88/062 (for municipal discharges), and shall include, at a minimum:

- i. Actions that will be taken to investigate/identify the causes/sources of toxicity,

- ii. Actions that will be evaluated to mitigate the impact of the discharge, to correct the non-compliance, and/or to prevent the recurrence of acute or chronic toxicity (this list of action steps may be expanded, if a TRE is undertaken), and
- iii. A schedule under which these actions will be implemented.

When monitoring measures toxicity in the effluent above a limitation established by this Order, the Discharger shall resample immediately, if the discharge is continuing, and retest for whole effluent toxicity. Results of an initial failed test and results of subsequent monitoring shall be reported to the Executive Officer (EO) as soon as possible following receipt of monitoring results. The EO will determine whether to initiate enforcement action, whether to require the Discharger to implement a Toxicity Reduction Evaluation, or to implement other measures. The Discharger shall conduct a TRE giving due consideration to guidance provided by the U.S. EPA's Toxicity Reduction Evaluation Procedures, Phases 1, 2, and 3 (EPA document Nos. EPA 600/3-88/034, 600/3-88/035, and 600/3-88/036, respectively). A TRE, if necessary, shall be conducted in accordance with the following schedule.

Table 9. Toxicity Reduction Evaluation—Schedule

Action Step	When Required
Take all reasonable measures necessary to immediately reduce toxicity, where the source is known.	Within 24 hours of identification of noncompliance.
Initiate the TRE in accordance to the Workplan.	Within 7 days of notification by the EO
Conduct the TRE following the procedures in the Workplan.	Within the period specified in the Workplan (not to exceed one year, without an approved Workplan)
Submit the results of the TRE, including summary of findings, required corrective action, and all results and data.	Within 60 days of completion of the TRE
Implement corrective actions to meet Permit limits and conditions.	To be determined by the EO

3. Best Management Practices and Pollution Prevention

This section of the standardized permit template is not applicable.

4. Construction, Operation and Maintenance Specifications

This section of the standardized permit template is not applicable.

5. Special Provisions for Municipal Facilities (POTWs Only)

- a. **Biosolids Management.** The handling, management, and disposal of sludge and solids derived from wastewater treatment must comply with applicable provisions of USEPA regulations at 40 CFR 257, 258, 501, and 503, including all monitoring, record keeping, and reporting requirements.

Solids and sludge treatment, storage, and disposal or reuse shall not create a nuisance, such as objectionable odors or flies, and shall not result in groundwater contamination. Sites for solids and sludge treatment and storage shall have adequate facilities to divert surface water runoff from adjacent areas to protect the boundaries of such sites from erosion, and to prevent drainage from treatment and storage sites.

The treatment, storage, disposal, or reuse of sewage sludge and solids shall not cause waste material to be in a position where it is, or can be, conveyed from the treatment and storage sites and deposited into waters of the State. The Discharger is responsible for assuring that all biosolids produced at its facility are used or disposed of in accordance with the above rules, whether the Discharger uses or disposes of the biosolids itself, or transfers them to another party for further treatment, use, or disposal. The Discharger is responsible for informing subsequent preparers, applicers, and disposers of the requirements that they must adhere to under these rules.

- b. Pretreatment Requirements.** The Discharger shall be responsible for the performance of all pretreatment requirements contained in 40 CFR 403 and shall be subject to enforcement actions, penalties, fines, and other remedies by the USEPA, or other appropriate parties, as provided in the CWA, as amended (33 USA 1351 et seq.). The Discharger shall implement and enforce its Approved Publicly Owned Treatment Works (POTW) Pretreatment Program. Implementation of the Discharger's Approved POTW Pretreatment Program is hereby made an enforceable condition of this permit. USEPA may initiate enforcement action against an industrial user for non-compliance with applicable standards and requirements as provided in the CWA.

The Discharger shall enforce the requirements promulgated under Sections 307 (b), (c), & (d) and 402 (b) of the CWA. The Discharger shall cause industrial users subject to Federal Categorical Standards to achieve compliance no later than the date specified in those requirements or, in the case of a new industrial user, upon commencement of the discharge.

The Discharger shall perform the pretreatment functions as required in 40 CFR Part 403, including, but not limited to:

- i. Implement necessary legal authorities as provided in 40 CFR 403.8 (f)(1);
- ii. Enforce the pretreatment requirements under 40 CFR 403.5 and 403.6;
- iii. Implement the programmatic functions as provided in 40 CFR 403.8 (f)(2); and,
- iv. Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR 403.8 (f)(3).

The Discharger shall submit annually a report to the USEPA - Region 9, the Regional Board, and the State Water Resources Control Board describing the

Discharger's pretreatment activities over the previous twelve months. In the event that the Discharger is not in compliance with conditions or requirements of this permit affected by the pretreatment program, it shall also include reasons for non-compliance and a statement how and when it shall comply. This annual report is due by January 30 of each year and shall contain.

The Discharger shall comply, and ensure affected "indirect dischargers" comply with Paragraph No. II.D.1 of the "Standard Provisions and Reporting Requirements".

6. Other Special Provisions

- a. **Discharges of Storm Water.** For the control of storm water discharged from the site of the wastewater treatment facilities, if applicable, the Discharger shall seek authorization to discharge under and meet the requirements of the State Water Resources Control Board's Water Quality Order 97-03-DWQ, NPDES General Permit No. CAS000001, *Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities Excluding Construction Activities*.
- b. **Statewide General Waste Discharge Requirements for Sanitary Sewer Systems (State Water Board Order No. 2006-0003-DWQ).** This General Permit, adopted on May 2, 2006, is applicable to all "federal and State agencies, municipalities, counties, districts, and other public entities that own or operate sanitary sewer systems greater than one mile in length that collect and/or convey untreated or partially treated wastewater to a publicly owned treatment facility in the State of California." The purpose of the General Permit is to promote the proper and efficient management, operation, and maintenance of sanitary sewer systems and to minimize the occurrences and impacts of sanitary sewer overflows. If applicable, the Discharger shall seek coverage under the General Permit and comply with its requirements.

7. Compliance Schedules

This section of the standardized permit template is not applicable.

8. Salt & Nutrient Management Program

- a. The Discharger shall maintain an ongoing salt/nutrient management program with the intent of reducing mass loading of salts and nutrients (with an emphasis on nitrogen species) in treated effluent to a level that will ensure compliance with effluent limitations and protect beneficial uses of groundwater.
- b. Salt reduction measures shall focus on all potential salt contributors to the collection system, including water supply, commercial, industrial and residential dischargers. The salt/nutrient management program shall also address the concentration of salts in the wastewater treatment process as a result of excessive hydraulic retention times and/or chemical addition.

- c. Nutrient reduction measures shall focus on optimizing wastewater treatment processes for nitrification and denitrification, or other means of nitrogen removal. Reduction measures may also include source control (non-human waste from commercial and industrial sources) as appropriate.
- d. As part of the salt/nutrient management program, the Discharger shall submit an annual report of salt and nutrient reduction efforts. This salt/nutrient management report shall be included as part of the annual report described in Monitoring and Reporting Program No. R3-2010-0009. The report shall be submitted by January 30th, and shall include (at a minimum):

Salt Component

- i. Calculations of annual salt mass discharged to (influent) and from (effluent) the wastewater treatment or recycling facility with analysis of contributing sources;
- ii. Analysis of wastewater evaporation/salt concentration effects;
- iii. Analysis of groundwater monitoring results related to salt constituents;
- iv. Analysis of potential adverse effects of salt loading on the groundwater basin;
- v. A summary of existing salt reduction measures; and,
- vi. Recommendations and time schedules for implementation of any additional salt reduction measures.

Nutrient Component

- i. Calculations of annual nitrogen mass (for all identified species) discharged to (influent) and from (effluent) the wastewater treatment or recycling facility with analysis of contributing sources;
 - ii. Analysis of groundwater monitoring results related to nitrogen constituents;
 - iii. Analysis of potential impacts of nitrogen loading on the groundwater basin;
 - iv. A summary of existing nitrogen loading reduction measures; and,
 - v. Recommendations and time schedules for implementation of any additional nitrogen loading reduction measures.
- e. As an alternative to the salt/nutrient management program requirements described above, upon Executive Officer approval, the Discharger may submit documentation and summary of participation in a regional salt/nutrient management plan implemented under the provisions of State Water Board Resolution No. 2009-0011 (Recycled Water Policy).

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in section IV of this Order will be determined as specified below.

A. General

Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined in the MRP and Attachment A of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).

B. Multiple Sample Data

When determining compliance with an AMEL, AWEL, or MDEL for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

1. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
2. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

ATTACHMENT A – DEFINITIONS

Arithmetic Mean (μ)

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean} = \mu = \Sigma x / n \quad \text{where: } \Sigma x \text{ is the sum of the measured ambient water concentrations, and } n \text{ is the number of samples.}$$

Average Monthly Effluent Limitation (AMEL)

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL)

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Bioaccumulative

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

Carcinogenic

Pollutants are substances that are known to cause cancer in living organisms.

Coefficient of Variation (CV)

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

Daily Discharge

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

Detected, but Not Quantified (DNQ)

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL.

Dilution Credit

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

Effluent Concentration Allowance (ECA)

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in USEPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

Enclosed Bays

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

Estimated Chemical Concentration

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

Estuaries

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

Inland Surface Waters

All surface waters of the State that do not include the ocean, enclosed bays, or estuaries.

Instantaneous Maximum Effluent Limitation

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Maximum Daily Effluent Limitation (MDEL)

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

Median

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median = $X_{(n+1)/2}$. If n is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the $n/2$ and $n/2+1$).

Method Detection Limit (MDL)

MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in title 40 of the Code of Federal Regulations, Part 136, Attachment B, revised as of July 3, 1999.

Minimum Level (ML)

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Mixing Zone

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

Not Detected (ND)

Sample results which are less than the laboratory's MDL.

Ocean Waters

The territorial marine waters of the State as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. Discharges to ocean waters are regulated in accordance with the State Water Board's California Ocean Plan.

Persistent Pollutants

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

Pollutant Minimization Program (PMP)

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

Pollution Prevention

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State or Regional Water Board.

Reporting Level (RL)

RL is the ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in this Order. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Regional Water Board either from Appendix 4 of the SIP in accordance with section 2.4.2 of the SIP or established in accordance with section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

Satellite Collection System

The portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

Source of Drinking Water

Any water designated as municipal or domestic supply (MUN) in a Regional Water Board Basin Plan.

Standard Deviation (σ)

Standard Deviation is a measure of variability that is calculated as follows:

$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:

x is the observed value;

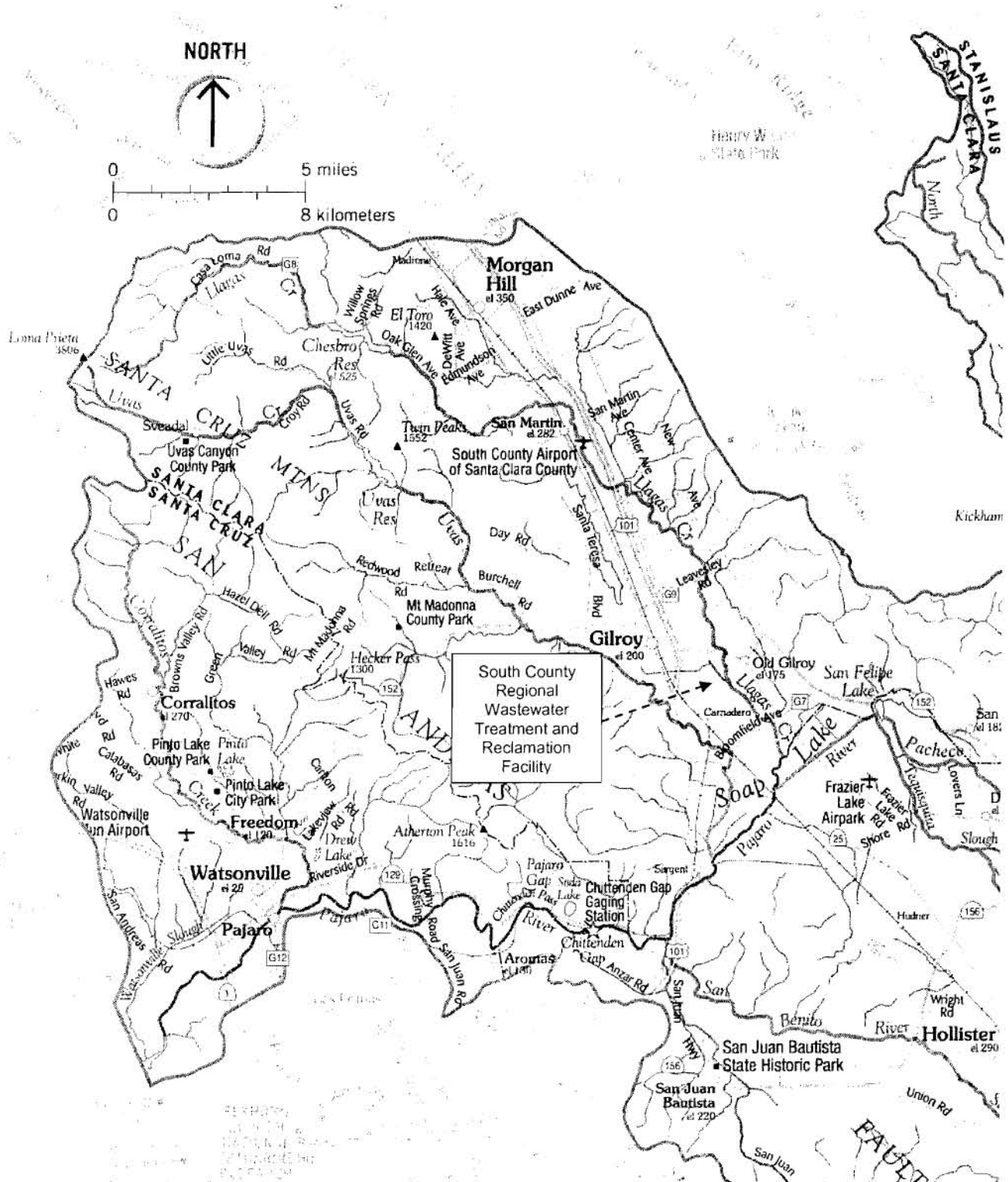
μ is the arithmetic mean of the observed values; and

n is the number of samples.

Toxicity Reduction Evaluation (TRE)

TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

ATTACHMENT B – MAP



Existing

- 1 Headworks
- 2 Solids Processing
- 3 Pre-anoxic Basins
- 4 Oxidation Ditches
- 5 Post Anoxic Basin
- 6 Recirculation Basin

Future

- 7 Secondary Clarifiers
- 8 Anthracite Filters
- 9 Chlorine Contact Basins
- 10 Administration Building
- 11 Secondary Effluent Pump Station
- 12 Filter Pump Station
- 13 Reclamation Pump Station
- 14 Southwest Reclamation Pump Station
- 15 UV Disinfection Basins
- 16 MBF Treatment Train

The diagram illustrates the wastewater treatment process flow. Wastewater enters the Headworks (1) and moves to Solids Processing (2). It then passes through Pre-anoxic Basins (3) and Oxidation Ditches (4). The flow continues to the Post Anoxic Basin (5) and the Recirculation Basin (6). From there, it goes to the Secondary Clarifiers (7) and Anthracite Filters (8). The water then moves through Chlorine Contact Basins (9) and the Administration Building (10). The Secondary Effluent Pump Station (11) and Filter Pump Station (12) are also shown. The Reclamation Pump Station (13) and Southwest Reclamation Pump Station (14) are located near the UV Disinfection Basins (15) and MBF Treatment Train (16). The final discharge points are labeled: To Storage Reservoir, To Percolation Ponds, To South Pipeline (River Discharge, Recycled Water), To North Reclamation Pipeline, and To Percolation Ponds.

ATTACHMENT D – STANDARD PROVISIONS

I. FEDERAL STANDARD PROVISIONS – PERMIT COMPLIANCE

A. Duty to Comply

1. The Discharger must comply with all of the conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification; or denial of a permit renewal application. (40 CFR § 122.41(a).)
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 CFR § 122.41(a)(1).)

B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 CFR § 122.41(c).)

C. Duty to Mitigate

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 CFR § 122.41(d).)

D. Proper Operation and Maintenance

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 CFR § 122.41(e).)

E. Property Rights

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 CFR § 122.41(g).)

2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of State or local law or regulations. (40 CFR § 122.5(c).)

F. Inspection and Entry

The Discharger shall allow the Regional Water Board, State Water Board, United States Environmental Protection Agency (USEPA), and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (40 CFR § 122.41(i); Wat. Code, § 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (40 CFR § 122.41(i)(1));
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (40 CFR § 122.41(i)(2));
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (40 CFR § 122.41(i)(3)); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (40 CFR § 122.41(i)(4).)

G. Bypass

1. Definitions
 - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 CFR § 122.41(m)(1)(i).)
 - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR § 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 CFR § 122.41(m)(2).)

3. Prohibition of bypass. Bypass is prohibited, and the Regional Water Board may take enforcement action against a Discharger for bypass, unless (40 CFR § 122.41(m)(4)(i)):
 - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 CFR § 122.41(m)(4)(i)(A));
 - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 CFR § 122.41(m)(4)(i)(B)); and
 - c. The Discharger submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 CFR § 122.41(m)(4)(i)(C).)
4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 CFR § 122.41(m)(4)(ii).)
5. Notice
 - a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. (40 CFR § 122.41(m)(3)(i).)
 - b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). (40 CFR § 122.41(m)(3)(ii).)

H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 CFR § 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 CFR § 122.41(n)(2).)

2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 CFR § 122.41(n)(3)):
 - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 CFR § 122.41(n)(3)(i));
 - b. The permitted facility was, at the time, being properly operated (40 CFR § 122.41(n)(3)(ii));
 - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 CFR § 122.41(n)(3)(iii)); and
 - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 CFR § 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 CFR § 122.41(n)(4).)

II. FEDERAL STANDARD PROVISIONS – PERMIT ACTION

A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 CFR § 122.41(f).)

B. Duty to Reapply

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 CFR § 122.41(b).)

C. Transfers

This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the Water Code. (40 CFR § 122.41(l)(3); § 122.61.)

III. FEDERAL STANDARD PROVISIONS – MONITORING

- A. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 CFR § 122.41(j)(1).)

- B.** Monitoring results must be conducted according to test procedures under Part 136 or, in the case of sludge use or disposal, approved under Part 136 unless otherwise specified in Part 503 unless other test procedures have been specified in this Order. (40 CFR § 122.41(j)(4); § 122.44(i)(1)(iv).)

IV. FEDERAL STANDARD PROVISIONS – RECORDS

- A.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by Part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Regional Water Board Executive Officer at any time. (40 CFR § 122.41(j)(2).)

B. Records of monitoring information shall include:

1. The date, exact place, and time of sampling or measurements (40 CFR § 122.41(j)(3)(i));
2. The individual(s) who performed the sampling or measurements (40 CFR § 122.41(j)(3)(ii));
3. The date(s) analyses were performed (40 CFR § 122.41(j)(3)(iii));
4. The individual(s) who performed the analyses (40 CFR § 122.41(j)(3)(iv));
5. The analytical techniques or methods used (40 CFR § 122.41(j)(3)(v)); and
6. The results of such analyses. (40 CFR § 122.41(j)(3)(vi).)

C. Claims of confidentiality for the following information will be denied (40 CFR § 122.7(b)):

1. The name and address of any permit applicant or Discharger (40 CFR § 122.7(b)(1)); and
2. Permit applications and attachments, permits and effluent data. (40 CFR § 122.7(b)(2).)

V. FEDERAL STANDARD PROVISIONS – REPORTING

A. Duty to Provide Information

The Discharger shall furnish to the Regional Water Board, State Water Board, or USEPA within a reasonable time, any information which the Regional Water Board, State Water Board, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance

with this Order. Upon request, the Discharger shall also furnish to the Regional Water Board, State Water Board, or USEPA copies of records required to be kept by this Order. (40 CFR § 122.41(h); Wat. Code, § 13267.)

B. Signatory and Certification Requirements

1. All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 CFR § 122.41(k).)
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA). (40 CFR § 122.22(a)(3).)
3. All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 CFR § 122.22(b)(1));
 - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 CFR § 122.22(b)(2)); and
 - c. The written authorization is submitted to the Regional Water Board and State Water Board. (40 CFR § 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Regional Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 CFR § 122.22(c).)
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

"I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure

that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 CFR § 122.22(d).)

C. Monitoring Reports

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 CFR § 122.22(l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. (40 CFR § 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under Part 136 or, in the case of sludge use or disposal, approved under Part 136 unless otherwise specified in Part 503, or as specified in this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Regional Water Board. (40 CFR § 122.41(l)(4)(ii).)
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 CFR § 122.41(l)(4)(iii).)

D. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 CFR § 122.41(l)(5).)

E. Twenty-Four Hour Reporting

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. (40 CFR § 122.41(l)(6)(i).)
2. The following shall be included as information that must be reported within 24 hours under this paragraph (40 CFR § 122.41(l)(6)(ii)):

- a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 CFR § 122.41(l)(6)(ii)(A).)
 - b. Any upset that exceeds any effluent limitation in this Order. (40 CFR § 122.41(l)(6)(ii)(B).)
3. The Regional Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 CFR § 122.41(l)(6)(iii).)

F. Planned Changes

The Discharger shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 CFR § 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 CFR § 122.41(l)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are subject neither to effluent limitations in this Order nor to notification requirements under section 122.42(a)(1) (see Additional Provisions—Notification Levels VII.A.1). (40 CFR § 122.41(l)(1)(ii).)
3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan. (40 CFR § 122.41(l)(1)(iii).)

G. Anticipated Noncompliance

The Discharger shall give advance notice to the Regional Water Board or State Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with General Order requirements. (40 CFR § 122.41(l)(2).)

H. Other Noncompliance

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. (40 CFR § 122.41(l)(7).)

I. Other Information

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, State Water Board, or USEPA, the Discharger shall promptly submit such facts or information. (40 CFR § 122.41(l)(8).)

VI. FEDERAL STANDARD PROVISIONS – ENFORCEMENT

- A.** The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13385, 13386, and 13387

VII. FEDERAL ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

A. Publicly-Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Regional Water Board of the following (40 CFR § 122.42(b)):

1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 CFR § 122.42(b)(1)); and
2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 CFR § 122.42(b)(2).)
3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 CFR § 122.42(b)(3).)

ATTACHMENT D-1 - CENTRAL COAST WATER BOARD STANDARD PROVISIONS (JANUARY 1985)

I. Central Coast General Permit Conditions

A. Central Coast Standard Provisions – Prohibitions

1. Introduction of "incompatible wastes" to the treatment system is prohibited.
2. Discharge of high-level radiological waste and of radiological, chemical, and biological warfare agents is prohibited.
3. Discharge of "toxic pollutants" in violation of effluent standards and prohibitions established under Section 307(a) of the Clean Water Act is prohibited.
4. Discharge of sludge, sludge digester or thickener supernatant, and sludge drying bed leachate to drainageways, surface waters, or the ocean is prohibited.
5. Introduction of pollutants into the collection, treatment, or disposal system by an "indirect discharger" that:
 - a. Inhibit or disrupt the treatment process, system operation, or the eventual use or disposal of sludge; or,
 - b. Flow through the system to the receiving water untreated; and,
 - c. Cause or "significantly contribute" to a violation of any requirement of this Order, is prohibited.
6. Introduction of "pollutant free" wastewater to the collection, treatment, and disposal system in amounts that threaten compliance with this order is prohibited.

B. Central Coast Standard Provisions – Provisions

1. Collection, treatment, and discharge of waste shall not create a nuisance or pollution, as defined by Section 13050 of the California Water Code.
2. All facilities used for transport or treatment of wastes shall be adequately protected from inundation and washout as the result of a 100-year frequency flood.
3. Operation of collection, treatment, and disposal systems shall be in a manner that precludes public contact with wastewater.
4. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed in a manner approved by the Executive Officer.
5. Publicly owned wastewater treatment plants shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to CCR Title 23.

6. After notice and opportunity for a hearing, this order may be terminated for cause, including, but not limited to:
 - a. violation of any term or condition contained in this order;
 - b. obtaining this order by misrepresentation, or by failure to disclose fully all relevant facts;
 - c. a change in any condition or endangerment to human health or environment that requires a temporary or permanent reduction or elimination of the authorized discharge; and,
 - d. a substantial change in character, location, or volume of the discharge.
7. Provisions of this permit are severable. If any provision of the permit is found invalid, the remainder of the permit shall not be affected.
8. After notice and opportunity for hearing, this order may be modified or revoked and reissued for cause, including:
 - a. Promulgation of a new or revised effluent standard or limitation;
 - b. A material change in character, location, or volume of the discharge;
 - c. Access to new information that affects the terms of the permit, including applicable schedules;
 - d. Correction of technical mistakes or mistaken interpretations of law; and,
 - e. Other causes set forth under Sub-part D of 40 CFR Part 122.
9. Safeguards shall be provided to assure maximal compliance with all terms and conditions of this permit. Safeguards shall include preventative and contingency plans and may also include alternative power sources, stand-by generators, retention capacity, operating procedures, or other precautions. Preventative and contingency plans for controlling and minimizing the affect of accidental discharges shall:
 - a. identify possible situations that could cause "upset", "overflow" or "bypass", or other noncompliance. (Loading and storage areas, power outage, waste treatment unit outage, and failure of process equipment, tanks and pipes should be considered.)
 - b. evaluate the effectiveness of present facilities and procedures and describe procedures and steps to minimize or correct any adverse environmental impact resulting from noncompliance with the permit.
10. Physical Facilities shall be designed and constructed according to accepted engineering practice and shall be capable of full compliance with this order when properly operated and maintained. Proper operation and maintenance shall be

described in an Operation and Maintenance Manual. Facilities shall be accessible during the wet-weather season.

11. Production and use of reclaimed water is subject to the approval of the Board. Production and use of reclaimed water shall be in conformance with reclamation criteria established in Chapter 3, Title 22, of the CCR and Chapter 7, Division 7, of the California Water Code. An engineering report pursuant to section 60323, Title 22, CCR is required and a waiver of water reclamation requirements from the Board is required before reclaimed water is supplied for any use, or to any user, not specifically identified and approved either in this Order or another order issued by this Board.

C. Central Coast Standard Provisions – General Monitoring Requirements

1. If results of monitoring a pollutant appear to violate effluent limitations based on a weekly, monthly, 30-day, or six-month period, but compliance or non-compliance cannot be validated because sampling is too infrequent, the frequency of sampling shall be increased to validate the test within the next monitoring period. The increased frequency shall be maintained until the Executive Officer agrees the original monitoring frequency may be resumed.

For example, if copper is monitored annually and results exceed the six-month median numerical effluent limitation in the permit, monitoring of copper must be increased to a frequency of at least once every two months (Central Coast Standard Provisions – Definitions I.G.13.). If suspended solids are monitored weekly and results exceed the weekly average numerical limit in the permit, monitoring of suspended solids must be increased to at least four (4) samples every week (Central Coast Standard Provisions – Definitions I.G.14.).

2. Water quality analyses performed in order to monitor compliance with this permit shall be by a laboratory certified by the State Department of Health Services for the constituent(s) being analyzed. Bioassay(s) performed in order to monitor compliance with this permit shall be in accord with guidelines approved by the State Water Resources Control Board and the State Department of Fish and Game. If the laboratory used or proposed for use by the discharger is not certified by the California Department of Health Services or, where appropriate, the Department of Fish and Game due to restrictions in the State's laboratory certification program, the discharger shall be considered in compliance with this provision provided:
 - a. Data results remain consistent with results of samples analyzed by the Regional Water Board;
 - b. A quality assurance program is used at the laboratory, including a manual containing steps followed in this program that is available for inspections by the staff of the Regional Water Board; and,
 - c. Certification is pursued in good faith and obtained as soon as possible after the program is reinstated.

3. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. Samples shall be taken during periods of peak loading conditions. Influent samples shall be samples collected from the combined flows of all incoming wastes, excluding recycled wastes. Effluent samples shall be samples collected downstream of the last treatment unit and tributary flow and upstream of any mixing with receiving waters.
4. All monitoring instruments and devices used by the discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy.

E. Central Coast Standard Provisions – General Reporting Requirements

1. Reports of marine monitoring surveys conducted to meet receiving water monitoring requirements of the Monitoring and Reporting Program shall include at least the following information:
 - a. A description of climatic and receiving water characteristics at the time of sampling (weather observations, floating debris, discoloration, wind speed and direction, swell or wave action, time of sampling, tide height, etc.).
 - b. A description of sampling stations, including differences unique to each station (e.g., station location, grain size, rocks, shell litter, calcareous worm tubes, evident life, etc.).
 - c. A description of the sampling procedures and preservation sequence used in the survey.
 - d. A description of the exact method used for laboratory analysis. In general, analysis shall be conducted according to Central Coast Standard Provisions – C.1 above, and Federal Standard Provision – Monitoring III.B. However, variations in procedure are acceptable to accommodate the special requirements of sediment analysis. All such variations must be reported with the test results.
 - e. A brief discussion of the results of the survey. The discussion shall compare data from the control station with data from the outfall stations. All tabulations and computations shall be explained.
2. Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule shall be submitted within 14 days following each scheduled date unless otherwise specified within the permit. If reporting noncompliance, the report shall include a description of the reason, a description and schedule of tasks necessary to achieve compliance, and an estimated date for achieving full compliance. A second report shall be submitted within 14 days of full compliance.
3. The “Discharger” shall file a report of waste discharge or secure a waiver from the Executive Officer at least 180 days before making any material change or proposed change in the character, location, or plume of the discharge.

4. Within 120 days after the discharger discovers, or is notified by the Regional Water Board, that monthly average daily flow will or may reach design capacity of waste treatment and/or disposal facilities within four (4) years, the discharger shall file a written report with the Regional Water Board. The report shall include:
 - a. the best estimate of when the monthly average daily dry weather flow rate will equal or exceed design capacity; and,
 - b. a schedule for studies, design, and other steps needed to provide additional capacity for waste treatment and/or disposal facilities before the waste flow rate equals the capacity of present units.

In addition to complying with Federal Standard Provision – Reporting V.B., the required technical report shall be prepared with public participation and reviewed, approved and jointly submitted by all planning and building departments having jurisdiction in the area served by the waste collection, treatment, or disposal facilities.

5. All “Dischargers” shall submit reports to the:

California Regional Water Quality Control Board
Central Coast Region
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401-7906

In addition, “Dischargers” with designated major discharges shall submit a copy of each document to:

Regional Administrator
US Environmental Protection Agency, Region 9
Attention: CWA Standards and Permits Office (WTR-5)
75 Hawthorne Street
San Francisco, California 94105

6. Transfer of control or ownership of a waste discharge facility must be preceded by a notice to the Regional Water Board at least 30 days in advance of the proposed transfer date. The notice must include a written agreement between the existing “Discharger” and proposed “Discharger” containing specific date for transfer of responsibility, coverage, and liability between them. Whether a permit may be transferred without modification or revocation and reissuance is at the discretion of the Board. If permit modification or revocation and reissuance is necessary, transfer may be delayed 180 days after the Regional Water Board's receipt of a complete permit application. Please also see Federal Standard Provision – Permit Action II.C.
7. Except for data determined to be confidential under Section 308 of the Clean Water Act (excludes effluent data and permit applications), all reports prepared in accordance with this permit shall be available for public inspection at the office of the Regional Water Board or Regional Administrator of EPA. Please also see Federal Standard Provision – Records IV.C.

8. By January 30th of each year, the discharger shall submit an annual report to the Regional Water Board. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. The discharger shall discuss the compliance record and corrective actions taken, or which may be needed, to bring the discharge into full compliance. The report shall address operator certification and provide a list of current operating personnel and their grade of certification. The report shall inform the Board of the date of the Facility's Operation and Maintenance Manual (including contingency plans as described Central Coast Standard Provision – Provision B.9., above), of the date the manual was last reviewed, and whether the manual is complete and valid for the current facility. The report shall restate, for the record, the laboratories used by the discharger to monitor compliance with effluent limits and provide a summary of performance relative to Section C above, General Monitoring Requirements.

If the facility treats industrial or domestic wastewater and there is no provision for periodic sludge monitoring in the Monitoring and Reporting Program, the report shall include a summary of sludge quantities, analyses of its chemical and moisture content, and its ultimate destination.

If applicable, the report shall also evaluate the effectiveness of the local source control or pretreatment program using the State Water Resources Control Board's "Guidelines for Determining the Effectiveness of Local Pretreatment Programs."

F. Central Coast Standard Provisions – General Pretreatment Provisions

1. Discharge of pollutants by "indirect dischargers" in specific industrial sub-categories (appendix C, 40 CFR Part 403), where categorical pretreatment standards have been established, or are to be established, (according to 40 CFR Chapter 1, Subchapter N), shall comply with the appropriate pretreatment standards:
 - a. By the date specified therein;
 - b. Within three (3) years of the effective date specified therein, but in no case later than July 1, 1984; or,
 - c. If a new indirect discharger, upon commencement of discharge.

G. Central Coast Standard Provisions – Enforcement

1. Any person failing to file a report of waste discharge or other report as required by this permit shall be subject to a civil penalty not to exceed \$5,000 per day.
2. Upon reduction, loss, or failure of the treatment facility, the "Discharger" shall, to the extent necessary to maintain compliance with this permit, control production or all discharges, or both, until the facility is restored or an alternative method of treatment is provided.

H. Central Coast Standard Provisions – Definitions

(Not otherwise included in Attachment A to this Order)

1. A "composite sample" is a combination of no fewer than eight (8) individual samples obtained at equal time intervals (usually hourly) over the specified sampling (composite) period. The volume of each individual sample is proportional to the flow rate at the time of sampling. The period shall be specified in the Monitoring and Reporting Program ordered by the Executive Officer.
2. "Daily Maximum" limit means the maximum acceptable concentration or mass emission rate of a pollutant measured during a calendar day or during any 24-hour period reasonably representative of the calendar day for purposes of sampling. It is normally compared with results based on "composite samples" except for ammonia, total chlorine, phenolic compounds, and toxicity concentration. For all exceptions, comparisons will be made with results from a "grab sample".
3. "Discharger", as used herein, means, as appropriate: (1) the Discharger, (2) the local sewerage entity (when the collection system is not owned and operated by the Discharger), or (3) "indirect discharger" (where "Discharger" appears in the same paragraph as "indirect discharger", it refers to the discharger.)
4. "Duly Authorized Representative" is one where:
 - a. the authorization is made in writing by a person described in the signatory paragraph of Federal Standard Provision V.B.;
 - b. the authorization specifies either an individual or the occupant of a position having either responsibility for the overall operation of the regulated facility, such as the plant manager, or overall responsibility for environmental matters of the company; and,
 - c. the written authorization was submitted to the Regional Water Board.
5. A "grab sample" is defined as any individual sample collected in less than 15 minutes. "Grab samples" shall be collected during peak loading conditions, which may or may not be during hydraulic peaks. It is used primarily in determining compliance with the daily maximum limits identified in Central Coast Standard Provision – Provision G.2. and instantaneous maximum limits.
6. "Hazardous substance" means any substance designated under 40 CFR Part 116 pursuant to Section 311 of the Clean Water Act.
7. "Incompatible wastes" are:
 - a. Wastes which create a fire or explosion hazard in the treatment works;
 - b. Wastes which will cause corrosive structural damage to treatment works, but in no case wastes with a pH lower than 5.0 unless the works is specifically designed to accommodate such wastes;

- c. Solid or viscous wastes in amounts which cause obstruction to flow in sewers, or which cause other interference with proper operation of treatment works;
 - d. Any waste, including oxygen demanding pollutants (BOD, etc), released in such volume or strength as to cause inhibition or disruption in the treatment works and subsequent treatment process upset and loss of treatment efficiency; and,
 - e. Heat in amounts that inhibit or disrupt biological activity in the treatment works or that raise influent temperatures above 40°C (104°F) unless the treatment works is designed to accommodate such heat.
8. "Indirect Discharger" means a non-domestic discharger introducing pollutants into a publicly owned treatment and disposal system.
9. "Log Mean" is the geometric mean. Used for determining compliance of fecal or total coliform populations, it is calculated with the following equation:

$$\text{Log Mean} = (C_1 \times C_2 \times \dots \times C_n)^{1/n},$$

in which "n" is the number of days samples were analyzed during the period and any "C" is the concentration of bacteria (MPN/100 ml) found on each day of sampling. "n" should be five or more.

10. "Mass emission rate" is a daily rate defined by the following equations:

$$\text{mass emission rate (lbs/day)} = 8.34 \times Q \times C; \text{ and,}$$

$$\text{mass emission rate (kg/day)} = 3.79 \times Q \times C,$$

where "C" (in mg/L) is the measured daily constituent concentration or the average of measured daily constituent concentrations and "Q" (in MGD) is the measured daily flow rate or the average of measured daily flow rates over the period of interest.

11. The "Maximum Allowable Mass Emission Rate," whether for a month, week, day, or six-month period, is a daily rate determined with the formulas in paragraph G.10, above, using the effluent concentration limit specified in the permit for the period and the average of measured daily flows (up to the allowable flow) over the period.
12. "Maximum Allowable Six-Month Median Mass Emission Rate" is a daily rate determined with the formulas in Central Coast Standard Provision – Provision G.10, above, using the "six-month Median" effluent limit specified in the permit, and the average of measured daily flows (up to the allowable flow) over a 180-day period.
13. "Median" is the value below which half the samples (ranked progressively by increasing value) fall. It may be considered the middle value, or the average of two middle values.
14. "Monthly Average" (or "Weekly Average", as the case may be) is the arithmetic mean of daily concentrations or of daily mass emission rates over the specified 30-day (or 7-day) period.

$$\text{Average} = (X_1 + X_2 + \dots + X_n) / n$$

in which "n" is the number of days samples were analyzed during the period and "X" is either the constituent concentration (mg/l) or mass emission rate (kg/day or lbs/day) for each sampled day. "n" should be four or greater.

15. "Municipality" means a city, town, borough, county, district, association, or other public body created by or under State law and having jurisdiction over disposal of sewage, industrial waste, or other waste.
16. "Overflow" means the intentional or unintentional diversion of flow from the collection and transport systems, including pumping facilities.
17. "Pollutant-free wastewater" means inflow and infiltration, storm waters, and cooling waters and condensates which are essentially free of pollutants.
18. "Primary Industry Category" means any industry category listed in 40 CFR Part 122, Appendix A.
19. "Removal Efficiency" is the ratio of pollutants removed by the treatment unit to pollutants entering the treatment unit. Removal efficiencies of a treatment plant shall be determined using "Monthly averages" of pollutant concentrations (C, in mg/l) of influent and effluent samples collected about the same time and the following equation (or its equivalent):

$$C_{\text{Effluent}} \text{ Removal Efficiency (\%)} = 100 \times (1 - C_{\text{effluent}} / C_{\text{influent}})$$

20. "Severe property damage" means substantial physical damage to property, damage to treatment facilities which causes them to become inoperable, or substantial and permanent loss to natural resources which can reasonably be expected to occur in the absence of a "bypass". It does not mean economic loss caused by delays in production.
21. "Sludge" means the solids, residues, and precipitates separated from, or created in, wastewater by the unit processes of a treatment system.
22. To "significantly contribute" to a permit violation means an "indirect discharger" must:
 - a. Discharge a daily pollutant loading in excess of that allowed by contract with the "Discharger" or by Federal, State, or Local law;
 - b. Discharge wastewater which substantially differs in nature or constituents from its average discharge;
 - c. Discharge pollutants, either alone or in conjunction with discharges from other sources, which results in a permit violation or prevents sewage sludge use or disposal; or
 - d. Discharge pollutants, either alone or in conjunction with pollutants from other sources that increase the magnitude or duration of permit violations.

23. "Toxic Pollutant" means any pollutant listed as toxic under Section 307 (a) (1) of the Clean Water Act or under 40 CFR Part 122, Appendix D. Violation of maximum daily discharge limitations are subject to 24-hour reporting (Federal Standard Provisions V.E.).
24. "Zone of Initial Dilution" means the region surrounding or adjacent to the end of an outfall pipe or diffuser ports whose boundaries are defined through calculation of a plume model verified by the State Water Resources Control Board.

ATTACHMENT E – MONITORING AND REPORTING PROGRAM

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

NPDES regulations at 40 CFR 122.48 require that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Water Board to require technical and monitoring reports. This MRP establishes monitoring and reporting requirements, which implement the federal and California regulations.

I. GENERAL MONITORING PROVISIONS

- A. Laboratories analyzing monitoring samples shall be certified by the Department of Health Services, in accordance with Water Code section 13176, and must include quality assurance/quality control data with their reports.
- B. Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge. All samples shall be taken at the monitoring locations specified below and, unless otherwise specified, before the monitored flow joins or is diluted by any other waste stream, body of water, or substance. Monitoring locations shall not be changed without notification to and approval of the Regional Board.
- C. Appropriate flow measurement devices and methods consistent with accepted scientific practices shall be selected and used to ensure the accuracy and reliability of measurements of the volume of monitored discharges. The devices shall be installed, calibrated, and maintained to ensure that the accuracy of the measurements is consistent with the accepted capability of that type of device. Devices selected shall be capable of measuring flows with a maximum deviation of less than ± 10 percent from true discharge rates throughout the range of expected discharge volumes. Guidance in selection, installation, calibration, and operation of acceptable flow measurement devices can be obtained from the following references.
 - 1. A Guide to Methods and Standards for the Measurement of Water Flow, U.S. Department of Commerce, National Bureau of Standards, NBS Special Publication 421, May 1975, 96 pp. (Available from the U.S. Government Printing Office, Washington, D.C. 20402. Order by SD Catalog No. C13.10:421.)
 - 2. Water Measurement Manual, U.S. Department of Interior, Bureau of Reclamation, Second Edition, Revised Reprint, 1974, 327 pp. (Available from the U.S. Government Printing Office, Washington D.C. 20402. Order by Catalog No. 172.19/2:W29/2, Stock No. S/N 24003-0027.)
 - 3. Flow Measurement in Open Channels and Closed Conduits, U.S. Department of Commerce, National Bureau of Standards, NBS Special Publication 484, October 1977, 982 pp. (Available in paper copy or microfiche from National Technical Information Services (NTIS) Springfield, VA 22151. Order by NTIS No. PB-273 535/5ST.)
 - 4. NPDES Compliance Sampling Manual, U.S. Environmental Protection Agency, Office of Water Enforcement, Publication MCD-51, 1977, 140 pp. (Available from the General Services Administration (8FFS), Centralized Mailing Lists Services, Building 41, Denver Federal Center, CO 80225.)

- D. All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. All flow measurement devices shall be calibrated at least once per year to ensure continued accuracy of the devices.
- E. Monitoring results, including noncompliance, shall be reported at intervals and in a manner specified in this MRP.
- F. Unless otherwise specified by this MRP, all monitoring shall be conducted according to test procedures established at 40 CFR 136, *Guidelines Establishing Test Procedures for Analysis of Pollutants*. All analyses shall be conducted using the lowest practical quantitation limit achievable using the specified methodology. Where effluent limitations are set below the lowest achievable quantitation limits, pollutants not detected at the lowest practical quantitation limits will be considered in compliance with effluent limitations. Analysis for toxics listed by the California Toxics Rule shall also adhere to guidance and requirements contained in the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (2005).

II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order.

Table E-1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
---	INF-001	Influent wastewater at the plant headworks, prior to treatment and following all significant input of wastewater to the treatment system
001	EFF-001	Secondary-treated wastewater at a point located after all secondary treatment and prior to discharge to the percolation ponds
002	EFF-002	Tertiary-treated wastewater at a point after all treatment and prior to contact with the receiving water
---	SW-001	A location at the drain on the south side of East Ponds at culvert prior to entering Llagas Creek
---	SW-002	A location at the drain on the north side of South Ponds at culvert prior to entering Llagas Creek
---	SW-003A	A location at the farm drainage east of the East Ponds
---	SW-004	A location at the drain at the southeast corner of Pond S9 that drains to Llagas Creek
---	SW-005A	A location at the outlet of Miller Slough near Luchessa Avenue bridge
---	SW-006A	A location at the outlet of the city storm drain near Luchessa Avenue bridge
---	SW-007	A location in Llagas Creek 600 feet north of Bloomfield Road
---	SW-008	A location in Llagas Creek 637 feet north of the northwest corner of Pond E1 and south of Highway 152
---	SW-009	A location in Llagas Creek 1000 feet north of Highway 152
---	SW-010	A location in Llagas Creek on the north side of Bloomfield Road bridge

---	RSW-011	A location in Pajaro River 100 feet upstream of Discharge Point 002
---	RSW-012	A location in Pajaro River 100 feet downstream of Discharge Point 002
---	GW-001	Groundwater well MW1/PWA located within the municipal ponds
---	GW-002	Groundwater well PWF located within the municipal ponds
---	GW-003	Groundwater well MW13 located within the municipal ponds
---	GW-004	Groundwater well MW24 located within the municipal ponds
---	GW-005	Groundwater well MW2/PWB located within the municipal ponds
---	GW-006	Groundwater well MW7/PWT located within the municipal ponds
---	GW-007	Groundwater well PWX located within the municipal ponds
---	GW-008	Groundwater well PWC located within the municipal ponds
---	GW-009	Groundwater well MW12/PWV located within the municipal ponds
---	GW-010	Groundwater well PWY located within the municipal ponds
---	GW-011	Groundwater well MW3/PWD located within the former food process ponds
---	GW-012	Groundwater well MW6/PWH located within the former food process ponds
---	GW-013	Groundwater well MW4/PWK located within the former food process ponds
---	GW-014	Groundwater well MW5/PWN located within the former food process ponds
---	GW-015	Groundwater well PWR located within the former food process ponds
---	GW-016	Groundwater well PWE located within the former food process ponds
---	GW-017	Groundwater well PWI located within the former food process ponds
---	GW-018	Groundwater well PWL located within the former food process ponds
---	GW-019	Groundwater well PWP located within the former food process ponds
---	GW-020	Groundwater well PWS located within the former food process ponds
---	GW-021	Groundwater well MW8/PWG located within the former food process ponds
---	GW-022	Groundwater well MW9/PWJ located within the former food process ponds
---	GW-023	Groundwater well PWM located within the former food process ponds
---	GW-024	Groundwater well PWQ located within the former food process ponds
---	GW-025	Groundwater well MW10 located north of Southside Drive
---	GW-026	Groundwater well MW21 located east of the railway line
---	GW-027	Groundwater well MW26 located within the municipal ponds
---	BIO-001	Biosolids at the last point in the biosolids handling process where representative samples of residual solids from the treatment process can be obtained

III. INFLUENT MONITORING REQUIREMENTS

A. Monitoring Location INF-001

1. The Discharger shall monitor influent to the facility at INF-001 as follows:

Table E-2. Influent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency
Flow ^[1]	MGD	Metered	Continuously
TSS ^[2]	mg/L	24-hr Composite	Every 8 th day
BOD ₅ ^[2]	mg/L	24-hr Composite	Every 8 th day
Specific Conductance	µmhos/cm	Metered	Continuously

^[1] Flow reporting shall include mean daily flow, maximum daily flow, and average monthly flow.

^[2] Collection of TSS and BOD₅ influent samples shall occur on days that effluent samples are collected.

IV. EFFLUENT MONITORING REQUIREMENTS

A. Monitoring Location EFF-002

1. The Discharger shall monitor tertiary effluent at monitoring location EFF-002, as follows. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level. Discrete discharge periods shall be defined by lapses in discharge flow of 24 hours or more. When discrete discharges occur at Discharge Point 002, monitoring shall occur at least once during the first discrete discharge period of the sampling period. Monitoring for CTR pollutants shall occur once during the permit term, during discharge to the Pajaro River, in the 12 months period before application is made to renew WDRs for the facility. Monitoring results at EFF-001 for the same parameters may be used to satisfy monitoring requirements established for EFF-002.

Table E-3. Effluent Monitoring – EFF-002

Parameter	Units	Sample Type	Minimum Sampling Frequency ^[1]
Flow ^[2]	MGD	Metered	Continuous
BOD ₅	mg/L	Grab	Every 8 th Day
TSS	mg/L	Grab	Every 8 th Day
Settleable Solids	mL/L	Grab	Every 8 th Day
pH ^{[3][4]}	s.u.	Grab	Daily
Chlorine Used	lbs/day	Calculated	Daily
Chlorine Residual	mg/L	Metered	Continuous
Modal Contact Time	minutes	Metered/Calc	Continuous
Dissolved Oxygen	mg/L	Grab	Weekly
Turbidity	NTU	Metered	Continuous
Fecal Coliform Bacteria	MPN/100 mL	Grab	Weekly
Total Coliform Bacteria	MPN/100mL	Grab	Weekly
Temperature ^[3]	° F	Grab	Daily

Parameter	Units	Sample Type	Minimum Sampling Frequency ^[1]
Color	Color Units	Grab	Monthly
Un-ionized Ammonia ^[3]	mg/L as N	Calculation	Every 8 th Day
Total Ammonia ^[3]	mg/L as N	Grab	Every 8 th Day
Nitrate ^[4]	mg/L as N	Grab	Every 8 th Day
Total Kjeldahl Nitrogen	mg/L as N	Grab	Monthly
Acute Toxicity ^[5]	Pass or Fail	24-hr composite	Annually
Chronic Toxicity ^[6]	TUc	24-hr composite	Annually
TDS ^[4]	mg/L	Grab	Monthly
Sodium ^[4]	mg/L	Grab	Monthly
Chloride ^[4]	mg/L	Grab	Monthly
Sulfate ^[4]	mg/L	Grab	Monthly
Boron ^[4]	mg/L	Grab	Monthly
Aluminum	mg/L	Grab	Monthly
Manganese ^[4]	mg/L	Grab	Monthly
Alkalinity ^[4]	mg/L	Grab	Monthly
mg/L	Grab	Monthly	
Calcium ^[4]	mg/L	Grab	Monthly
Carbonate ^[4]	mg/L	Grab	Monthly
Copper ^[4]	µg/L	24-hr composite	Monthly
Electrical Conductivity ^[4]	µg/L	Grab	Monthly
Fluoride ^[4]	mg/L	Grab	Monthly
Iron ^[4]	mg/L	Grab	Monthly
Magnesium ^[4]	mg/L	Grab	Monthly
Nitrite ^[4]	mg/L	Grab	Monthly
Potassium ^[4]	mg/L	Grab	Monthly
Zinc ^[4]	mg/L	Grab	Monthly
Lead	µg/L	24-hr composite	Monthly
Chlorodibromomethane ^[5]	µg/L	24-hr composite	Monthly
CTR Pollutants ^{[7] [8] [9]}	µg/L	24-hr composite	1X / Permit Term
2,3,7,8-TCDD Equivalents ^{[8][9]}	µg/L	24-hr composite	1X / Permit Term
Title 22 Pollutants ^{[10] [11]}	µg/L	24-hr composite	Quarterly

^[1] Monthly and quarterly monitoring shall occur on the first 8th day sample of the month. Quarterly sampling shall occur in March, June, September, and October.

^[2] Flow reporting shall include mean daily flow, maximum daily flow, and average monthly flow.

^[3] Temperature and pH are to be measured at the same time the Total Ammonia sample is collected. Results shall be used to calculate and report Unionized Ammonia concentrations.

^[4] General Mineral and Irrigation Suitability (except MBAS) pollutant list.

^[5] Whole effluent acute toxicity monitoring shall be conducted according to the requirements established in Section V.A of this Monitoring and Reporting Plan.

^[6] Whole effluent chronic toxicity monitoring shall be conducted according to the requirements established in Section V.B of this Monitoring and Reporting Plan.

^[7] Those 126 pollutants with applicable water quality objectives established by the California Toxics Rule (CTR) at 40 CFR 131.38.

^[8] Analyses, compliance determination, and reporting for these pollutants shall adhere to applicable provisions of the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (SIP). The Discharger shall instruct its analytical laboratory to establish calibration standards so that the Minimum Levels (MLs) presented in Appendix 4 of the SIP are the lowest calibration standards. The Discharger and its analytical laboratory shall select MLs that are below applicable water

- quality criteria of the CTR; and when applicable water quality criteria are below all MLs, the Discharger and its analytical laboratory shall select the lowest ML.
- [9] 24-hour composite samples shall be collected one time, during discharge, and within the twelve-month period before application is made to renew the Waste Discharge Requirements for the facility.
- [10] The Title 22 pollutants are those pollutants for which the Department of Health Services has established Maximum Contaminant Levels (MCLs) at Title 22, Division 4, Chapter 15, sections 64431 (Inorganic Chemicals), 64444 (Organic Chemicals), and 64442 and 64443 (Radionuclides) of the California Code of Regulations.
- [11] Analytical methods shall adhere to the Detection Limits for Purposes of Reporting (DLRs) established by Title 22 of the California Code of Regulations, Division 4, Chapter 15, section 64432 (Inorganics), section 64445.1 (Organics), and sections 64442 and 64443 (Radionuclides).

V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

A. Whole Effluent Acute Toxicity

1. Acute Toxicity Monitoring Requirements - EFF-002

- a. Bioassays shall be performed to evaluate the toxicity of the discharge in accordance with the following procedures unless otherwise specified by the Water Board's Executive Officer or designee:
- b. Both test species given below shall be used to measure acute toxicity:

Table E-4. Approved Test for Acute Toxicity

Species	Effect	Test Duration (days)	Reference
Fathead Minnow (<i>Pimephales promelas</i>)	Larval Survival and Growth	7	EPA/821-R-02-012 (Acute)
Water Flea (<i>Ceriodaphnia dubia</i>)	Survival and Reproduction	7	EPA/821-R-02-012 (Acute)

- c. Determination of acute toxicity shall be based on mortality data derived from chronic toxicity tests, utilizing these species, as specified below.
- d. The presence of acute toxicity shall be determined as significantly reduced survival of test organisms at 100 percent effluent compared to a control using a statistical t-test.

B. Whole Effluent Chronic Toxicity

1. Chronic Toxicity Monitoring Requirements – EFF-002

- a. *Sampling.* The Discharger shall collect 24-hour composite samples of the effluent at Discharge Point 002, when discharge to the Pajaro River is occurring, for critical life stage toxicity testing as indicated below. For toxicity tests requiring renewals, 24-hour composite samples collected on consecutive days are required.
- b. *Test Species.* The Discharger shall utilize the water flea, *Ceriodaphnia dubia*, (survival and reproduction test); fathead minnow, *Pimephales promelas* (larval survival and growth test); and green alga, *Selenastrum capricornutum* (growth

test), as test species. The Executive Officer may change to another test species if data suggest that another test species is more sensitive to the discharge.

- c. *Methodology.* Sample collection, handling and preservation shall be in accordance with USEPA protocols. In addition, bioassays shall be conducted in compliance with the most recently promulgated test methods, currently "Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms," Fourth Edition (EPA-821-R-02-013), with exceptions granted the Discharger by the Executive Officer and the Environmental Laboratory Accreditation Program (ELAP).
- d. *Dilution Series.* The Discharger shall conduct toxicity testing at 100% effluent.

2. Chronic Toxicity Reporting Requirements

- a. *Routine Reporting.* Toxicity test results for the current reporting period shall include, at a minimum, for each test:
 - (1) Sample date(s)
 - (2) Test initiation date
 - (3) Test species
 - (4) End point values for each dilution (e.g., number of young, growth rate, percent survival)
 - (5) NOEC value(s) in percent effluent
 - (6) IC15, IC25, IC40, and IC50 values (or EC15, EC25 ... etc.) as percent effluent
 - (7) TUc values (100/NOEC, 100/IC25, or 100/EC25)
 - (8) Mean percent mortality ($\pm$ s.d.) after 96 hours in 100% effluent (if applicable)
 - (9) NOEC and LOEC values for reference toxicant test(s)
 - (10) IC50 or EC50 value(s) for reference toxicant test(s)
 - (11) Available water quality measurements for each test (pH, D.O., temperature, conductivity, hardness, salinity, ammonia)
- b. *Compliance Summary.* The results of the chronic toxicity testing shall be provided in the self-monitoring report and shall include a summary table of chronic toxicity data from at least eleven of the most recent samples. The information in the table shall include items listed above under 2.a, specifically item numbers 1, 3, 5, 6(IC25 or EC25), 7, and 8.

C. Quality Assurance

1. The use of a dilution series for this Discharger is not applicable, because there is no dilution in the receiving water.
2. For the acute toxicity testing using a t-test, two dilutions shall be used, i.e., 100 percent effluent and a control (when a t-test is used instead of an LC50).
3. If organisms are not cultured in-house, concurrent testing with a referenced toxicant shall be conducted. Where organisms are cultured in-house, monthly reference toxicant testing is sufficient. Reference toxicant tests also shall be conducted using the same test conditions as the effluent toxicity tests (e.g., same test duration, etc.).
4. If either the reference toxicant test or effluent test does not meet all test acceptability criteria (TAC) as specified in the toxicity test references, then the permittee must re-sample and retest within 15 working days or as soon as possible. The retesting period begins when the Discharger collects the first sample required to complete the retest.
5. The reference toxicant and effluent tests must meet the upper and lower bounds on test sensitivity as determined by calculating the percent minimum significant difference (PMSD) for each test result. The test sensitivity bound is specified for each test method in the respective methods manuals.

D. Accelerated Monitoring Requirements

1. When acute toxicity is detected in the effluent, or when a chronic toxicity effluent trigger value of 1 TUc is exceeded during regular toxicity monitoring, and the testing meets all test acceptability criteria, the Discharger shall initiate accelerated monitoring to confirm the effluent toxicity.
2. The Discharger shall implement an accelerated monitoring frequency consisting of performing three toxicity tests in a six-week period following the first failed test results.
3. If implementation of the generic Toxicity Reduction Evaluation (TRE) work plan indicates the source of the exceedance of the toxicity trigger (for instance, a temporary plant upset), then only one additional test is necessary. If exceedance of the toxicity trigger is detected in this test, the Discharger will continue with accelerated monitoring requirements or implement the Toxicity Identification and Toxicity Reduction Evaluations.
4. If none of the three tests indicated exceedance of the toxicity trigger, then the Discharger may return to the normal bioassay testing frequency.

E. Conducting Toxicity Identification Evaluations and Toxicity Reduction Evaluations

1. A Toxicity Identification Evaluation (TIE) shall be triggered if testing from the accelerated monitoring frequency indicates any of the following:

- a. Two of the three accelerated toxicity tests are reported as failed tests meeting any of the conditions specified in Attachment E, Section V.D.
 - b. The TIE shall be initiated within 15 days following failure of the second accelerated monitoring test.
 - c. If a TIE is triggered prior to the completion of the accelerated testing, the accelerated testing schedule may be terminated, or used as necessary in performing the TIE.
2. The TIE shall be conducted to identify and evaluate toxicity in accordance with procedures recommended by the United States Environmental Protection Agency (USEPA) which include the following:
 - a. Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents, Phase I, (USEPA, 1992a);
 - b. Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures, Second Edition (USEPA, 1991a);
 - c. Methods for Aquatic Toxicity Identification Evaluations: Phase II Toxicity Identification Procedures for Sampling Exhibiting Acute and Chronic Toxicity (USEPA, 1993a); and
 - d. Methods for Aquatic Toxicity Identification Evaluations: Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity (USEPA, 1993b).
3. As part of the TIE investigation, the Discharger shall be required to implement its TRE work plan. The Discharger shall take all reasonable steps to control toxicity once the source of the toxicity is identified. A failure to conduct required toxicity tests or a TRE within a designated period shall result in the establishment of numerical effluent limitations for chronic toxicity in a permit or appropriate enforcement action. Recommended guidance in conducting a TRE include the following:
 - a. Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants, August 1999, EPA/833B-99/002; and
 - b. Clarifications Regarding Toxicity Reduction and Identification Evaluations in the National Pollutant Discharge Elimination System Program dated March 27, 2001, USEPA Office of Wastewater Management, Office of Regulatory Enforcement.

VI. LAND DISCHARGE MONITORING REQUIREMENTS

A. Monitoring Location EFF-001

1. The Discharger shall monitor secondary effluent that is to be land applied at monitoring location EFF-001 as follows. If more than one analytical test method is

listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level. Results for monitoring requirements at EFF-001 may be used to satisfy monitoring requirements at EFF-002 for duplicative parameters.

Table E-5. Land Discharge Monitoring – EFF-001

Parameter	Units	Sample Type	Minimum Sampling Frequency ⁽¹⁾
Flow ^[2]	MGD	Metered	Continuously
Settleable Solids	mL/L	Grab	Weekly
pH ^[3]	s.u.	Grab	Daily
BOD ₅	mg/L	24-hr composite	Every 8 th Day
TSS	mg/L	24-hr composite	Every 8 th Day
Total Ammonia	mg/L as N	Grab	Monthly
Nitrate ^[3]	mg/L as N	Grab	Monthly
Total Kjeldahl Nitrogen	mg/L as N	Grab	Monthly
TDS ^[3]	mg/L	Grab	Quarterly
Sodium ^[3]	mg/L	Grab	Quarterly
Chloride ^[3]	mg/L	Grab	Quarterly
Sulfate ^[3]	mg/L	Grab	Quarterly
Alkalinity ^[3]	mg/L	Grab	Quarterly
Bicarbonate ^[3]	mg/L	Grab	Quarterly
Boron ^[3]	mg/L	Grab	Quarterly
Calcium ^[3]	mg/L	Grab	Quarterly
Carbonate ^[3]	mg/L	Grab	Quarterly
Electrical Conductivity ^[3]	µmhos/cm	Grab	Quarterly
Fluoride ^[3]	mg/L	Grab	Quarterly
Iron ^[3]	mg/L	Grab	Quarterly
Magnesium ^[3]	mg/L	Grab	Quarterly
Manganese ^[3]	mg/L	Grab	Quarterly
Nitrite ^[3]	mg/L as N	Grab	Quarterly
Potassium ^[3]	mg/L	Grab	Quarterly
Zinc ^[3]	µg/L	Grab	Quarterly
Copper ^[3]	µg/L	Grab	Quarterly
Title 22 Pollutants ^{[4] [5]}	µg/L	24-hr composite	Annually

[1] Monthly and quarterly monitoring shall occur on the first 8th day sample of the month. Quarterly sampling shall occur in March, June, September, and October.

[2] Flow reporting shall include mean daily flow, maximum daily flow, and average monthly flow.

[3] General Mineral and Irrigation Suitability (except MBAS) pollutant list.

[4] The Title 22 pollutants are those pollutants for which the Department of Health Services has established Maximum Contaminant Levels (MCLs) at CCR Title 22, Division 4, Chapter 15, sections 64431 (Inorganic Chemicals) 64444 (Organic Chemicals), and 64442 and 64443 (Radionuclides).

[5] Analytical methods shall adhere to the Detection Limits for Purposes of Reporting (DLRs) established by CCR Title 22, Division 4, Chapter 15, section 64432 (Inorganics), section 64445.1 (Organics), and sections 64442 and 64443 (Radionuclides).

VII. RECLAMATION MONITORING REQUIREMENTS

This section of the standardized MRP is not applicable to the Discharger, as its Master Reclamation Requirements are contained in existing Order No. 98-052.

VIII. RECEIVING WATER MONITORING REQUIREMENTS – SURFACE WATER AND GROUNDWATER

A. Surface Water Monitoring Locations SW-001, SW-002, SW-003A, SW-004, SW-005A, SW-006A, SW-007, SW-008, SW-009, SW-010, RSW-011, and RSW-012.

1. The Discharger shall monitor surface waters at Monitoring Locations SW-001, SW-002, SW-003A, SW-004, SW-005A, SW-006A, SW-007, SW-008, SW-009, and SW-010, as follows.

Table E-6. Surface Water Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency ⁽¹⁾
Flow ⁽²⁾	MGD	Metered	Quarterly
Chemical Oxygen Demand	mg/L	Grab	Quarterly
Nitrate	mg/L as N	Grab	Quarterly
Total Ammonia	mg/L as N	Grab	Quarterly
TDS	mg/L	Grab	Quarterly
Sodium	mg/L	Grab	Quarterly
Chloride	mg/L	Grab	Quarterly
Dissolved Oxygen	mg/L	Grab	Quarterly
Temperature	°F	Grab	Quarterly
pH	s.u.	Grab	Quarterly
Turbidity	NTU	Grab	Quarterly
Fecal Coliform	MPN/100mL	Grab	Quarterly

⁽¹⁾ Quarterly monitoring shall occur in March, June, September, and December.

⁽²⁾ Flow reporting shall include maximum daily flow, mean daily flow, and average monthly flow. If no flow meter or gauging station exists, flow rate shall be estimated as accurately as possible.

2. The Discharger shall monitor receiving surface waters at Monitoring Locations RSW-011 and RSW-012, as follows. Monitoring shall occur concurrently with effluent monitoring at EFF-002. Receiving water monitoring requirements apply only when discharge to Pajaro River is occurring, except for flow and temperature, for determination of acceptable discharge periods. Discrete discharge periods are defined by lapses in discharge flows of 24 hours or more. When discrete discharges occur at Discharge Point 002, receiving water monitoring shall occur at least once during the first discrete discharge period of the sampling period.

Table E-7. Receiving Water Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency ⁽¹⁾
Flow ⁽²⁾	MGD	Metered	Prior to each discharge to Pajaro River, and daily during discharge

BOD ₅	mg/L	Grab	Monthly
TSS	mg/L	Grab	Monthly
Nitrate ^[3]	mg/L as N	Grab	Monthly
Total Ammonia ^[4]	mg/L as N	Grab	Monthly
Unionized Ammonia ^[4]	mg/L as N	Grab	Monthly
Total Kjeldahl Nitrogen	mg/L as N	Grab	Monthly
TDS ^[3]	mg/L	Grab	Monthly
Sodium ^[3]	mg/L	Grab	Monthly
Chloride ^[3]	mg/L	Grab	Monthly
Sulfate ^[3]	mg/L	Grab	Monthly
Boron ^[3]	mg/L	Grab	Monthly
Aluminum	mg/L	Grab	Monthly
Manganese ^[3]	mg/L	Grab	Monthly
Dissolved Oxygen	mg/L	Grab	Weekly
Temperature ^[4]	°F	Grab	Prior to each discharge to Pajaro River, and hourly during the discharge period and peak diurnal temperature period
pH ^{[3][4]}	s.u.	Grab	Weekly
Turbidity	NTU	Grab	Weekly
Fecal Coliform	MPN/100mL	Grab	Weekly
Alkalinity ^[3]	mg/L	Grab	Quarterly
Bicarbonate ^[3]	mg/L	Grab	Quarterly
Calcium ^[3]	mg/L	Grab	Quarterly
Carbonate ^[3]	mg/L	Grab	Quarterly
Copper ^[3]	mg/L	Grab	Quarterly
Electrical Conductivity ^[3]	µg/L	Grab	Quarterly
Fluoride ^[3]	mg/L	Grab	Quarterly
Iron ^[3]	mg/L	Grab	Quarterly
Magnesium ^[3]	mg/L	Grab	Quarterly
Nitrite ^[3]	mg/L	Grab	Quarterly
Potassium ^[3]	mg/L	Grab	Quarterly
Zinc ^[3]	mg/L	Grab	Quarterly
CTR Pollutants ^{[5][6][7]}	µg/L	24-hr composite	1X / Permit Term
2,3,7,8-TCDD Equivalents ^{[6][7]}	µg/L	24-hr composite	1X / Permit Term
Title 22 Pollutants ^{[7][8][9]}	µg/L	24-hr composite	1X / Permit Term

^[1] Quarterly monitoring shall occur in March, June, September, and December

^[2] Flow reporting shall include maximum daily flow, mean daily flow, and average monthly flow. Upstream flow shall be determined at the gauging station near the point of discharge. Downstream flow shall be determined at the USGS Chittenden gauging station.

^[3] General Mineral and Irrigation Suitability (except MBAS) pollutant list

^[4] Temperature and pH are to be measured at the same time the Total Ammonia sample is collected. Results shall be used to calculate and report Unionized Ammonia concentrations.

^[5] Those 126 pollutants with applicable water quality objectives established by the California Toxics Rule (CTR) at 40 CFR 131.38.

^[6] Analyses, compliance determination, and reporting for these pollutants shall adhere to applicable provisions of the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (SIP). The Discharger shall instruct its analytical laboratory to establish calibration standards so that

the Minimum Levels (MLs) presented in Appendix 4 of the SIP are the lowest calibration standards. The Discharger and its analytical laboratory shall select MLs, which are below applicable water quality criteria of the CTR; and when applicable water quality criteria are below all MLs, the Discharger and its analytical laboratory shall select the lowest ML.

- [7] 24-hour composite samples shall be collected one time, during discharge, and within the twelve-month period before application is made to renew the Waste Discharge Requirements for the facility.
- [8] The Title 22 pollutants are those pollutants for which the Department of Health Services has established Maximum Contaminant Levels (MCLs) at CCR Title 22, Division 4, Chapter 15, sections 64431 (Inorganic Chemicals) and 64444 (Organic Chemicals).
- [9] Analytical methods shall adhere to the Detection Limits for Purposes of Reporting (DLRs) established by Title 22 of the CCR, Division 4, Chapter 15, section 64432 (Inorganics) and section 64445.1 (Organics).

B. Monitoring Locations GW-001 through GW-027

1. The Discharger shall monitor groundwater at Monitoring Locations GW-001 through GW-027 as follows. Prior to sampling, wells shall be purged until dissolved oxygen levels, pH, and electrical conductivity have stabilized.

Table E-8. Groundwater Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency ^[1]
Groundwater Elevation ^[2]	feet	Observation	Quarterly
pH ^{[3][4]}	s.u.	Grab	[5]
Electrical Conductivity ^{[3][4]}	µmhos/cm	Grab	[5]
Alkalinity ^{[3][4]}	mg/L	Grab	[5]
Bicarbonate ^{[3][4]}	mg/L	Grab	[5]
Boron ^{[3][4]}	mg/L	Grab	[5]
Calcium ^{[3][4]}	mg/L	Grab	[5]
Carbonate ^{[3][4]}	mg/L	Grab	[5]
Chloride ^{[3][4]}	mg/L	Grab	[5]
Copper ^{[3][4]}	mg/L	Grab	[5]
Fluoride ^{[3][4]}	mg/L	Grab	[5]
Iron ^{[3][4]}	mg/L	Grab	[5]
Magnesium ^{[3][4]}	mg/L	Grab	[5]
Manganese ^{[3][4]}	mg/L	Grab	[5]
Nitrate ^{[3][4]}	mg/L	Grab	[5]
Nitrite ^{[3][4]}	mg/L	Grab	[5]
Potassium ^{[3][4]}	mg/L	Grab	[5]
Sodium ^{[3][4]}	mg/L	Grab	[5]
Sulfate ^{[3][4]}	mg/L	Grab	[5]
TDS ^{[3][4]}	mg/L	Grab	[5]
Zinc ^{[3][4]}	mg/L	Grab	[5]

[1] Quarterly monitoring shall occur in March, June, September, and December.

[2] Applicable to Monitoring Locations GW-001 through GW-024.

[3] General Mineral and Irrigation Suitability (except MBAS) pollutant list.

[4] Applicable only to Monitoring Locations GW-003, GW-004, GW-005, GW-006, GW-011, GW-013, GW-014, GW-020, GW-025, GW-026, and GW-027.

[5] Monitoring Locations GW-003, GW-004, GW-011, GW-014, and GW-025 shall be monitored quarterly in March, June, September, and December.
Monitoring Locations GW-005, GW-006, GW-013, GW-020, GW-026, and GW-027 shall be monitored bi-annually.

2. Groundwater monitoring reports shall include a site map showing the locations of all monitoring wells.

IX. OTHER MONITORING REQUIREMENTS

A. Biosolids Monitoring, Reporting, and Notification – BIO-001

1. A representative sample of biosolids shall be obtained from the last point in the handling process (i.e., in the drying beds just prior to removal or from a pond bottom). All constituents shall be analyzed annually for total concentrations for comparison with Total Threshold Limit Concentration (TTL) criteria. The Waste Extraction Test shall be performed on any constituent when the total concentration of the waste exceeds ten times the Soluble Threshold Limit Concentration Limit Concentration (STLC) limit for that substance. Twelve (12) discrete representative samples shall be collected at separate locations in the biosolids ready for disposal. These 12 samples shall be composited to form one (1) sample for constituent analysis. For accumulated, previously untested biosolids, the Discharger shall develop a representative sampling plan including number and location of sampling points, and collect representative samples. The analysis shall test for the metals required in 40 CFR 503.16 (for land application) or 503.26 (for surface disposal), using the methods in *Test Methods for Evaluating Solid Waste, Physical/Chemical Methods* (EPA Publication SW-846, all applicable editions and updates), as required in 40 CFR 503.8(b)(4), at the minimum frequencies established therein, provided in the table below.

Table E-9. Amount of Biosolids and Frequency for Analysis

Amount ^[1] (dry metric tons/ 365-day period)	Frequency ^[2]
Greater than zero, but less than 290	Once per year
Equal to or greater than 290 but less than 1500	Once per quarter (four times per year)
Equal to or greater than 1500 but less than 15,000	Once per sixty days (six times per year)
Greater than 15,000	Once per month (twelve times per year)

^[1] For land application, either the amount of bulk biosolids applied to the land or the amount prepared for sale or give-away in a bag or other container for application to the land (dry weight basis). If the Discharger's biosolids are directly land applied without further treatment by another preparer, biosolids shall also be tested for organic-N, ammonium-N, and nitrate-N at the frequencies required. For surface disposal, the amount of biosolids placed on an active sludge unit (dry weight basis).

^[2] Test results shall be expressed in mg pollutant per kg biosolids on a 100% dry weight basis.

Biosolids shall be analyzed for the constituents in the following table.

Table E-10. Biosolids Monitoring

Constituent	Units	Type of Sample	Sampling/Analysis Frequency
Quantity Removed	Tons or yds ³	Measured during Removal	Continual
Location of Reuse/Disposal	General Public or Specific Site	---	---

Moisture Content	%	Grab	Prior to reclamation/disposal of biosolids ^[1]
pH	Standard Units	Grab	Prior to reclamation/disposal of biosolids ^[1]
Total Kjeldahl Nitrogen	mg/kg (dry) ^[2]	Grab	Prior to reclamation/disposal of biosolids ^[1]
Ammonia(N)	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Nitrate(N)	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Total Phosphorus	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Grease and Oil	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Arsenic	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Antimony	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Barium	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Beryllium	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Boron	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Cadmium	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Cobalt	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Copper	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Chromium (total)	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Lead	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Mercury	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Molybdenum	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Nickel	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Selenium	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Silver	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Thallium	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Tin	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Vanadium	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Zinc	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Pesticides	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
Organic Lead	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]
PCBs	mg/kg	Grab	Prior to reclamation/disposal of biosolids ^[1]

^[1] Annually if sludge solids are being reclaimed or disposed of in that year.

^[2] Total sample (including solids and any liquid portion) to be analyzed and results reported as mg/kg based on the dry weight of the sample.

2. Prior to land application, the Discharger shall demonstrate that the biosolids meet Class A or Class B pathogen reduction levels by one of the methods listed in 40 CFR 503.32 (unless transferred to another preparer who demonstrates pathogen reduction.) Prior to disposal in a surface disposal site, the Discharger shall demonstrate that the biosolids meet Class B levels or shall ensure that the site is covered at the end of each operating day. If pathogen reduction is demonstrated using a "Process to Significantly/Further Reduce Pathogens" (PFRP), the Discharger shall maintain daily records of the operating parameters to achieve this reduction.

The following applies when biosolids from the Discharger are directly land applied as Class B, without further treatment by a second preparer. If the Discharger demonstrates pathogen reduction by direct testing for fecal coliforms and/or pathogens, samples must be drawn at the frequency in the Amount/Frequency table above. If the Discharger demonstrates Class B pathogen reduction by testing for

fecal coliform, at least seven grab samples must be drawn and analyzed during each monitoring event, and a geometric mean calculated from these seven samples. If the Discharger demonstrates Class A pathogen reduction by testing for fecal coliform and/or salmonella, plus one of the PFRP processes or testing for enteric viruses and helminth ova at least four samples of fecal coliform or salmonella must be drawn during each monitoring event. All four samples must meet the limits specified in 40 CFR 503.32(a).

3. For biosolids that are land applied or placed in a surface disposal site, the Discharger shall track and keep records of the operational parameters used to achieve Vector Attraction Reduction requirements in 40 CFR 503.33(b).
4. Class 1 facilities (facilities with pretreatment programs or others designated as Class 1 by the Regional Administrator) and Federal facilities with greater than five million gallons per day (MGD) influent flow shall sample biosolids for pollutants listed under Section 307(a) of the Clean Water Act (as required in the pretreatment section of the permit for POTWs with pretreatment programs). Class 1 facilities and Federal facilities greater than 5 MGD shall test dioxins/dibenzofurans using a detection limit of less than one pg/g at the time of their next priority pollutant scan if they have not done so within the past five years, and once per five years thereafter.
5. The biosolids shall be tested annually, or more frequently if necessary, to determine hazardousness. All constituents regulated under CA Title 22, division 5, chapter 11, article 3 shall be analyzed for comparison with TTLC criteria. The Waste Extraction Test shall be performed on any constituent when the total concentration of the waste exceeds ten times the STLC limit for that substance.
6. If biosolids are placed in a surface disposal site (dedicated land disposal site or monofill), a qualified groundwater scientist shall develop a groundwater monitoring program for the site, or shall certify that the placement of biosolids on the site will not contaminate an aquifer.
7. Biosolids placed in a municipal landfill shall be tested by the Paint Filter Liquids Test (EPA Method 9095) at the frequency determined by Table E-9, or more often if necessary to demonstrate that there are no free liquids.
8. The Discharger, either directly or through contractual agreements with their biosolids management contractors, shall comply with the following notification requirements:
 - a. *Notification of non-compliance.* The Discharger shall notify USEPA Region 9, the Regional Water Board, and the Regional Board located in the region where the biosolids are used or disposed, of any non-compliance within 24 hours if the non-compliance may seriously endanger health or the environment. For other instances of non-compliance, the Discharger shall notify USEPA Region 9 and the affected Regional Water Quality Boards of any non-compliance in writing within five working days of becoming aware of the non-compliance. The Discharger shall require their biosolids management contractors to notify USEPA Region 9 and the affected Regional Water Quality Boards of any non-compliance within the same time frames.

- b. If biosolids are shipped to another State or Indian lands, the Discharger must send notice at least 60 days prior to the shipment to the permitting authorities in the receiving State or Indian land (the USEPA Regional Office for that area and the State/Indian authorities).
- c. *For land application (in cases where Class B biosolids are directly applied without further treatment):* Prior to reuse of any biosolids from the Discharger's facility to a new or previously unreported site, the Discharger shall notify USEPA, the Regional Water Board, and any other affected Regional Water Quality Board. The notification shall include description of the crops or vegetation to be grown, proposed loading rates and determination of agronomic rates.

If any biosolids within a given monitoring period do not meet 40 CFR 503.13 metals concentrations limits, the Discharger (or its contractor) must pre-notify USEPA, and determine the cumulative metals loading to that site to date, as required in 40 CFR 503.12. The Discharger shall notify the applier of all the applier's requirements under 40 CFR 503, including the requirement that the applier certify that the management practices, site restrictions, and any applicable vector attraction reduction requirements have been met. The Discharger shall require the applier to certify at the end of 38 months following application of Class B biosolids that the harvesting restrictions in effect for up to 38 months have been met.

- d. *For surface disposal:* Prior to disposal to a new or previously unreported site, the Discharger shall notify USEPA and the Water Board. The notice shall include a description and a topographic map of the proposed site, depth to groundwater, whether the site is lined or unlined, site operator, site owner, and any State or local permits. The notice shall describe procedures for ensuring public access and grazing restrictions for three years following site closure. The notice shall include a groundwater monitoring plan or description of why groundwater monitoring is not required.
9. The Discharger shall submit an annual biosolids report to USEPA Region 9 Biosolids Coordinator and the Regional Board by February 19th of each year (per USEPA guidance and 40 CFR 503) for the period covering the previous calendar year. This report shall include:
- a. Annual biosolids removed in dry tons and percent solids.
 - b. If appropriate, a narrative description of biosolids dewatering and other treatment processes, including process parameters, including a schematic diagram showing biosolids handling facilities. For example, if drying beds are used, report depth of application and drying time. If composting is used, report the temperature achieved and duration.
 - c. A description of disposal methods, including the following information as applicable related to the disposal methods used at the facility. If more than one method is used, include the percentage and tonnage of annual biosolids production disposed by each method.

- (1) For landfill disposal include: 1) the Water Board WDR numbers that regulate the landfills used, 2) the present classifications of the landfills used, 3) the results of any groundwater monitoring, 4) certifications of management practices, and 5) the names and locations of the facilities receiving biosolids.
 - (2) For land application include: 1) the location of the site(s), 2) the Water Board's WDR numbers that regulate the site(s), 3) the application rate in lbs/acre/year (specify wet or dry), 4) certifications of management practices and site restrictions, and 5) subsequent uses of the land.
 - (3) For offsite application by a licensed hauler and composter include: 1) the name, address and USEPA license number of the hauler and composter.
- d. Copies of analytical data required by other agencies (i.e. USEPA or County Health Department) and licensed disposal facilities (i.e. landfill, land application, or composting facility) for the previous year.
 - e. Descriptions of pathogen reduction methods and vector attraction reduction methods. Including supporting time and temperature data, and certifications, as required in 40 CFR 503.17 and 503.27.
 - f. Names, mailing address, and street addresses of persons who received biosolids for storage, further treatment, disposal in a municipal waste landfill, or for other use or disposal methods not covered above, and amounts delivered to each.
 - g. For all biosolids used or disposed at the Discharger's facility, the site and management practice information and certification required in 40 CFR 503.17 and 503.27.
 - h. For all biosolids temporarily stored, the information required in 40 CFR 503.20 is required to demonstrate temporary storage.
 - i. Reports shall be submitted to:

Regional Biosolids Coordinator
USEPA (WTR-7)
75 Hawthorne St.
San Francisco, CA 94105-3901

Executive Officer
Central Coast Regional Water Quality Control Board
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401-7906

B. Pond Maintenance

The Discharger shall report on pond conditioning work conducted in the previous year with a summary included in the facility's Annual Report, due January 30th of each year. The

summary shall also contain a description of any problems encountered in operation of the system during the reporting period.

X. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

1. The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.

B. Self Monitoring Reports (SMRs)

1. At any time during the term of this permit, the State or Water Board may notify the Discharger to electronically submit Self-Monitoring Reports (SMRs) using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). Until such notification is given, the Discharger shall submit hard copy SMRs. The CIWQS web site will provide additional directions for SMR submittal in the event there will be service interruption for electronic submittal.
2. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. The Discharger shall submit monthly, quarterly, and annual SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule.

Table E-11. Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Continuous	Permit effective date	All	30 th day of the month following the reporting period
Daily	Permit effective date	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	30 th day of the month following the reporting period
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	30 th day of the month following the reporting period
Every 8 th Day	Sunday following permit effective date or on permit effective date if on a Sunday	14 day consecutive period	30 th day of the month following the reporting period

Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1 st day of calendar month through last day of calendar month	30 th day of the month following the reporting period
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit effective date	January 1 through March 31 April 1 through June 30 July 1 through September 30 October 1 through December 31	30 th day of the month following the reporting period
Annually	January 1 following (or on) permit effective date	January 1 through December 31	January 30
1x/permit term	January 1 following (or on) permit effective date	Permit term	180 days prior to permit expiration

4. Reporting Protocols. The Discharger shall report with each sample result the applicable reported Minimum Level (ML) and the current Method Detection Limit (MDL), as determined by the procedure in Part 136.

The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the reported ML shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ as well as the words "Estimated Concentration" (may be shortened to "Est. Conc."). The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy (+ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- c. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.
- d. Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.

5. Compliance Determination. Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined above and Attachment A of this Order. For purposes of reporting and administrative

enforcement by the Regional and State Water Boards, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).

6. Multiple Sample Data. When determining compliance with an AMEL, AWEL, or MDEL for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
 - a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
7. The Discharger shall submit SMRs in accordance with the following requirements:
 - a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.
 - b. The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.
 - c. SMRs must be submitted to the Water Board, signed and certified as required by the Standard Provisions (Attachment D), to the address listed below:

California Regional Water Quality Control Board
Central Coast Region
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401

C. Discharge Monitoring Reports (DMRs)

1. As described in Section X.B.1 above, at any time during the term of this permit, the State or Water Board may notify the Discharger to electronically submit SMRs that will satisfy federal requirements for submittal of Discharge Monitoring Reports (DMRs). Until such notification is given, the Discharger shall submit DMRs in accordance with the requirements described below.
2. DMRs must be signed and certified as required by the standard provisions (Attachment D). The Discharger shall submit the original DMR and one copy of the DMR to the address listed below:

STANDARD MAIL	FEDEX/UPS/ OTHER PRIVATE CARRIERS
State Water Resources Control Board Division of Water Quality c/o DMR Processing Center PO Box 100 Sacramento, CA 95812-1000	State Water Resources Control Board Division of Water Quality c/o DMR Processing Center 1001 I Street, 15 th Floor Sacramento, CA 95814

3. All discharge monitoring results must be reported on the official USEPA pre-printed DMR forms (EPA Form 3320-1). Forms that are self-generated will not be accepted unless they follow the exact same format of EPA Form 3320-1.

D. Other Reports

1. The Discharger shall report the results of any special monitoring, TREs, or other data or information that results from the Special Provisions, section VI. C, of the Order. The Discharger shall submit such reports with the first monthly SMR scheduled to be submitted on or immediately following the report due date.

ATTACHMENT F – FACT SHEET

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ATTACHMENT F – FACT SHEET

As described in section II of this Order, this Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

Table F-1. Facility Information

WDID	3 430100001
Discharger	South County Regional Wastewater Authority
Name of Facility	South County Wastewater Treatment and Reclamation Facility
Facility Address	1500 Southside Drive
	Gilroy, California 95020
	Santa Clara County
Facility Contact, Title and Phone	Saeid Vaziry, Chief Engineer, (408) 846-8842
Authorized Person to Sign and Submit Reports	Saeid Vaziry, Chief Engineer, (408) 846-8842
Mailing Address	7351 Rosanna Street, Gilroy, CA 95020
Billing Address	7351 Rosanna Street, Gilroy, CA 95020
Type of Facility	Publicly Owned Treatment Works (POTW)
Major or Minor Facility	Major
Threat to Water Quality	1
Complexity	B
Pretreatment Program	Y
Reclamation Requirements	Water Board Order No. 98-052
Facility Permitted Flow	8.5 million gallons per day (MGD) (average dry weather flow)
	10.2 MGD (average daily wet weather flow)
Facility Design Flow	8.5 MGD (average dry weather, secondary treatment capacity)
	9.0 MGD (tertiary treatment capacity)
Watershed	Pajaro River Watershed
Receiving Waters	Pajaro River
Receiving Water Type	Inland Surface Water

- A.** The South County Regional Wastewater Authority (SCRWA) is the owner of the South County Regional Treatment and Reclamation Facility, a municipal wastewater treatment plant operated by CH2M Hill OMI.

- B. For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and State laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.
- C. This Order authorizes the discharge of tertiary treated wastewater to the Pajaro River, waters of the United States, and it establishes waste discharge requirements for the land application of secondary treated wastewater. The Discharger also reclaims tertiary treated wastewater for irrigation and industrial use. If necessary, the terms and conditions of the current Order (Order No. R3-2004-0099) will be automatically continued past its expiration date of September 10, 2009, and remain in effect until new waste discharge requirements are adopted pursuant to this Order.
- D. The Discharger filed a Report of Waste Discharge and submitted an application for renewal of its Waste Discharge Requirements (WDRs) Order and National Pollutant Discharge Elimination System (NPDES) permit on March 9, 2009.

II. FACILITY DESCRIPTION

A. Description of Wastewater and Biosolids Treatment

The South County Wastewater Treatment and Reclamation Facility serves the Cities of Gilroy and Morgan Hill. The Cities of Gilroy and Morgan Hill own and maintain wastewater collections systems within each respective city. Wastewater from the two cities is conveyed to an interceptor sewer owned and maintained by the City of Gilroy and then to the SCRWA Reclamation Plant. As documented by Water Board Resolution No. R3-2008-0014, the facility has design, secondary treatment capacities of 8.5 million gallons per day (MGD, average dry weather flow) and 10.2 MGD (average wet weather flow) and a tertiary treatment capacity of 9.0 MGD. Secondary treated wastewater is land applied at Discharge Point 001 to percolation ponds located adjacent to Llagas Creek. Tertiary treated wastewater is reclaimed for irrigation use and may be discharged at Discharge Point 002 to the Pajaro River.

Secondary wastewater treatment is accomplished with two parallel treatment sequences, each including four pre-anoxic basins, an aerated oxidation ditch, and a secondary clarifier. A post-anoxic basin and a reaeration basin serve both sequences prior to discharge or delivery to tertiary treatment facilities which include filtration and chlorination/dechlorination steps.

Secondary treated wastewater is distributed from Discharge Point 001 to 37 percolation ponds, located adjacent to Llagas Creek, via a piped distribution network covering approximately 394 acres. The ponds consist of native agricultural land surrounded by dikes of compacted earth, characterized by clay and sandy clay soils. The pond system is run in irrigation mode, where effluent is applied approximately 6-12 inches deep and allowed to percolate completely, with a minimum of two days between applications to rest the soil. This practice also minimizes odors. Pond percolation capacities are estimated to be 10.5 MGD during the dry season and 9.2 MGD during the wet season. Ponds must be disked or plowed annually during the dry season to break up accumulated soils and keep the soils aerated.

Tertiary treated wastewater is used onsite for landscape irrigation and fire protection, however most tertiary treated wastewater is delivered offsite for agricultural use or industrial cooling by the City of Gilroy Parks, the Calpine Cogeneration Facility, the Gilroy Golf Course, the Eagle Ridge Golf Course, Obata Farms, and the McCarthy Business Park. Under emergency conditions during wet weather events, tertiary treated wastewater can also be discharged at Discharge Point 002 to the Pajaro River; however, during the 2004 – 2008 time period, there were no discharges at this outfall. During the anticipated five year term of this Order, ultraviolet disinfection capability and a new outfall line will be installed to manage discharges to the Pajaro River.

Biosolids are wasted five days per week, and dewatered using belt presses. Dewatered biosolids are placed in the Marina Landfill in Monterey County as an alternate daily cover.

Groundwater is present below the percolation ponds at a depth of approximately 5 – 25 feet below the surface. Groundwater elevations increase during the winter recharge season (November – March) and decrease during the summer pumping season (April – October).

B. Discharge Points and Receiving Waters

Secondary treated effluent is distributed at Discharge Point 001 (36° 58' 50" N, 121° 32' 00" W) for land application to 37 percolation ponds adjacent to Llagas Creek. Tertiary treated effluent is used primarily for Title 22-compliant reclamation use, and may be discharged intermittently to the Pajaro River at Discharge Point 002 (36° 57' 00" N, 121° 30' 43" W).

C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

Effluent limitations and land discharge specifications contained in the previous Order for secondary and tertiary effluent, and monitoring data collected during the permit term, are summarized in the table below.

Table F-2. Historic Effluent Limitations and Monitoring Data

Parameter	Units	Effluent Limitation				Monitoring Data (Sept 2004 – Dec 2008)	
		30-Day Mean	7-Day Mean	Maximum Daily	12-Month Moving Median	Maximum 30-Day Mean	Maximum 12-Month Moving Median
Secondary Effluent Limitations							
BOD ₅	mg/L	30	35	--	---	7	---
TSS	mg/L	30	45	--	---	8	---
Nitrate as N	mg/L	5	---	10	---	4.6	---
Total Dissolved Solids (TDS)	mg/L	---	---	---	900	---	825
Chloride	mg/L	---	---	---	200	---	188
Sodium	mg/L	---	---	---	175	---	123
Sulfate	mg/L	---	---	---	150	---	74
Boron	mg/L	---	---	---	1.0	---	1

Parameter	Units	Effluent Limitation				Monitoring Data (Sept 2004 – Dec 2008)	
		30-Day Mean	7-Day Mean	Maximum Daily	12-Month Moving Median	Maximum 30-Day Mean	Maximum 12-Month Moving Median
pH	s.u.	6.5 to 8.3 at all times			---	Minimum- 7.3 Maximum – 7.7	---
Tertiary Effluent Limitations							
BOD ₅	mg/L	10	---	20	---	No Discharge - No Effluent Data	
TSS	mg/L	10	---	20	---	"	
Nitrate as N	mg/L	5	---	10	---	"	
Unionized Ammonia as N	mg/L	0.025	---	0.050	---	"	
TDS	mg/L	1,000	---	---	---	"	
Chloride	mg/L	250	---	---	---	"	
Sodium	mg/L	200	---	---	---	"	
Sulfate	mg/L	250	---	---	---	"	
Boron	mg/L	1.0	---	---	---	"	
Lead	µg/L	2.12	---	4.26	---	"	
Thallium	µg/L	1.7	---	3.42	---	"	
Chloroform	µg/L	1.1	---	2.21	---	"	
Dibromochloromethane	µg/L	0.40	---	0.81	---	"	
Bromodichloromethane	µg/L	0.56	---	1.13	---	"	
Bis(2-ethylhexyl)phthalate	µg/L	1.8	---	3.62	---	"	
Aluminum	µg/L	1,000	---	2,010	---	"	
Manganese	µg/L	200	---	400	---	"	
Turbidity	NTU	- Daily average must be less than or equal to 2 NTU - Must be less than 10 NTU at all times - Must not exceed 5 NTU for more than 5 percent of the time			---	"	
Total Coliform Bacteria	MPN/100 mL	- Seven-day median shall not exceed 2.2/100 mL - Coliform concentrations shall not exceed 23/100 mL in more than one sample within a 30-day period - Any single sample shall not exceed 240/100 mL			---	"	
Residual Chlorine	mg/L	There shall not be measurable chlorine residual in the effluent			---	"	

D. Compliance Summary

The Discharger has been in compliance with all limitations and requirements of Order No. R3-2004-0099.

E. Planned Changes

Design and construction of an ultraviolet disinfection system and a new outfall pipeline with a capacity to convey up to 9 MGD of treated and disinfected effluent to the Pajaro River to manage extreme wet weather flows will be completed during the term of this Order.

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in the proposed Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and chapter 5.5, division 7 of the California Water Code (CWC) (commencing with section 13370). It shall serve as a NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the Water Code (commencing with section 13260).

B. California Environmental Quality Act (CEQA)

Pursuant to Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code sections 21100 - through 21177.

C. State and Federal Regulations, Policies, and Plans

- 1. Water Quality Control Plans.** The Water Board has adopted a *Water Quality Control Plan for the Central Coast Region* (the Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for receiving waters within the Region. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which establishes State policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Beneficial uses established by the Basin Plan for the Pajaro River are presented below.

Table F-3. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
002	Pajaro River	• Municipal and domestic water supply (MUN) • Agricultural supply (AGR) • Industrial process supply (PRO) • Groundwater recharge (GWR) • Contact (REC-1) and Non-contact (REC-2) water recreation • Wildlife habitat (WILD) • Cold freshwater habitat (COLD) • Warm freshwater habitat (WARM) • Migration of aquatic organisms (MIGR) • Spawning, reproduction, and/or early development (SPWN) • Freshwater replenishment (FRESH) • Commercial and sport fishing (COMM).

Beneficial uses established by the Basin Plan for groundwaters include municipal and domestic water supply, agricultural supply, and industrial process supply. Requirements of this Order implement the Basin Plan.

2. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, USEPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the State. The CTR was amended on February 13, 2001. These rules contain water quality criteria for priority pollutants that are applicable to the receiving waters for discharges from the facility.
3. **State Implementation Policy.** On March 2, 2000, the State Water Board adopted the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Water Board in the Basin Plan. The SIP became effective on May 18, 2000, with respect to the priority pollutant criteria promulgated by the USEPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control that are applicable to discharges to the receiving waters for discharges from the facility. Requirements of this Order implement the SIP.
4. **Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised state and tribal water quality standards become effective for CWA purposes [65 Fed. Reg. 24641 (April 27, 2000) (codified at 40 CFR 131.21,)]. Under the revised regulation (also known as the Alaska Rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect

and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.

5. **Antidegradation Policy.** NPDES regulations at 40 CFR 131.12 require that State water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16, which incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that the existing quality of waters be maintained unless degradation is justified based on specific findings. The Water Board's Basin Plan implements and incorporates by reference both the State and federal antidegradation policies. As discussed in section IV.D.2 of this Fact Sheet, the permitted discharge is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution No. 68-16.
6. **Anti-Backsliding Requirements.** CWA Sections 402 (o) (2) and 303 (d) (4) and NPDES regulations at 40 CFR 122.44 (l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. As discussed in section IV.D.1 of this Fact Sheet, effluent limitations and other requirements established by this Order satisfy applicable anti-backsliding provisions of the CWA and NPDES regulations.

D. Impaired Water Bodies on CWA 303(d) List

CWA section 303 (d) requires states to identify specific water bodies where water quality standards are not expected to be met after implementation of technology based limitations on point sources. For all 303 (d) listed water bodies, the Water Board must develop and implement TMDLs (total maximum daily loads) that specify WLAs (waste load allocations) for point sources and load allocations for non-point sources.

The State's 2006 303 (d) list of impaired water bodies, which was approved by USEPA in June 2007 identifies the Pajaro River as being impaired for boron and fecal coliform.

A Total Maximum Daily Load (TMDL) for fecal coliform bacteria for the Pajaro River watershed, which includes the Pajaro River and Llagas Creek, has been adopted by the Regional Water Board (Water Board Order No. R3-2009-0008.) TMDLs have also been adopted and approved by USEPA for sediment and nitrate for the Pajaro River watershed. The TMDL for fecal coliform prohibits all fecal coliform loading from human sources to the Pajaro River. The TMDL for nitrate finds that current actions of the Regional Water Board adequately implements the TMDL and will be adequate to correct the impairment due to nitrate. The TMDL for sediment requires no actions by SCRWA. TMDLs for the remaining impairing pollutants are anticipated to be developed by 2019. This Order includes requirements of all TMDLs that are applicable to the SCRWA Wastewater Treatment and Reclamation Facility.

E. Other Plans, Policies and Regulations

- 1. Statewide General Waste Discharge Requirements for Sanitary Sewer Systems (State Water Board Order No. 2006-0003-DWQ).** This General Permit, adopted on May 2, 2006, is applicable to all "federal and State agencies, municipalities, counties, districts, and other public entities that own or operate sanitary sewer systems greater than one mile in length that collect and/or convey untreated or partially treated wastewater to a publicly owned treatment facility in the State of California." The purpose of the General Permit is to promote the proper and efficient management, operation, and maintenance of sanitary sewer systems and to minimize the occurrences and impacts of sanitary sewer overflows. If applicable, the Discharger must seek coverage under the General Permit and comply with its requirements.
- 2. Discharges of Storm Water.** For the control of storm water discharged from the site of the wastewater treatment and disposal facilities, if applicable, the Order requires the Discharger to seek authorization to discharge under and meet the requirements of the State Water Resources Control Board's Water Quality Order 97-03-DWQ, NPDES General Permit No. CAS000001, *Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities Excluding Construction Activities*.

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. NPDES regulations establish two principal bases for effluent limitations. At 40 CFR 122.44 (a) permits are required to include applicable technology-based limitations and standards; and at 40 CFR 122.44 (d) permits are required to include water quality-based effluent limitations (WQBELs) to attain and maintain applicable numeric and narrative water quality criteria (WQC) to protect the beneficial uses of the receiving water. When numeric water quality objectives (WQOs) have not been established, but a discharge has the reasonable potential to cause or contribute to an excursion above a narrative criterion, WQBELs may be established using one or more of three methods described at 40 CFR 122.44 (d) - 1) WQBELs may be established using a calculated water quality criterion derived from a proposed State criterion or an explicit State policy or regulation interpreting its narrative criterion; 2) WQBELs may be established on a case-by-case basis using USEPA criteria guidance published under CWA Section 304 (a); or 3) WQBELs may be established using an indicator parameter for the pollutant of concern.

A. Discharge Prohibitions

- 1. Discharge Prohibition III. A** (No discharge at a location or in a manner except as described by the Order). The Order authorizes a single point of discharge of tertiary effluent to the Pajaro River, a single point of discharge for land application of secondary effluent, and a single point of discharge for recycled water distribution.

This prohibition reflects CWA section 402's prohibition against discharges of pollutants except in compliance with the Act's permit requirements, effluent limitations, and other enumerated provisions. This prohibition is also retained from the previous permit.

2. **Discharge Prohibition III. B** (The discharge of any waste not specifically regulated by this permit is prohibited.) Because limitations and conditions of the Order have been prepared based on specific information provided by the Discharger and specific wastes described by the Discharger, the limitations and conditions of the Order do not adequately address waste streams not contemplated during drafting of the Order. To prevent the discharge of such waste streams that may be inadequately regulated, the Order prohibits the discharge of any waste that was not described to the Water Board during the process of permit reissuance. This prohibition is retained from the previous permit.
3. **Discharge Prohibition III. C** (Creation of a condition of pollution, contamination, or nuisance, as defined by Section 13050 of the CWC, is prohibited). This prohibition is retained from the previous permit.
4. **Discharge Prohibition III. D** (Overflows and bypasses prohibited). The discharge of untreated or partially treated wastewater from the Discharger's collection, treatment, or disposal facilities represents an unauthorized bypass pursuant to 40 CFR 122.41 (m) or an unauthorized discharge, which poses a threat to human health and/or aquatic life, and therefore, is explicitly prohibited by the Order.
5. **Discharge Prohibition III. E** (Discharges of sludge to surface waters prohibited). This prohibition is retained from the previous permit, and is based on the solid waste discharge prohibition against the discharge of solids to surface waters contained in the Basin Plan at section VI.D.1 of Chapter 4.
6. **Discharge Prohibition III. F** (Average dry weather daily flow shall not exceed 8.5 MGD and average wet weather daily flow shall not exceed 10.2 MGD.) This prohibition is retained from the previous permit, where it was expressed as an effluent limitation. The purpose of the prohibition is to assure that influent flows do not exceed the treatment plant's design capacities, and thereby, to assure efficient treatment of wastewater.
7. **Discharge Prohibition III. G** (Discharge of fecal coliform bacteria at Discharge Point 002 to the Pajaro River is prohibited.) This prohibition is established by this Order to implement the TMDL for fecal coliform bacteria for the Pajaro River watershed, adopted by the Water Board through Order No. R3-2009-0008.

B. Technology-Based Effluent Limitations

1. Scope and Authority

NPDES regulations at 40 CFR 122.44 (a) require that permits include applicable technology-based limitations and standards. Where the USEPA has not yet developed technology based standards for a particular industry or a particular

pollutant, CWA Section 402 (a) (1) and USEPA regulations at 40 CFR 125.3 authorize the use of best professional judgment (BPJ) to derive technology-based effluent limitations on a case-by-case basis. When BPJ is used, the permit writer must consider specific factors outlined at 40 CFR 125.3.

At 40 CFR 133 in the Secondary Treatment Regulations, USEPA has established the following minimum required level of effluent quality attainable by secondary treatment.

Table F-4. Secondary Treatment Requirements

Parameter	30-Day Average	7-Day Average
BOD ^[1]	30 mg/L	45 mg/L
TSS ^[1]	30 mg/L	45 mg/L
pH	6.0 – 9.0 s.u.	

^[1] The 30-day average percent removal shall not be less than 85 percent.

2. Applicable Technology-Based Effluent Limitations

The following table summarizes technology-based effluent limitations established by the Order for the discharge to the Pájaro River at Discharge Point 002.

Table F-6. Summary of Technology-Based Effluent Limitations – Discharge Point 002

Parameter	Units	Effluent Limitations - Discharge Point 002	
		Average Monthly	Average Weekly
BOD ₅ ^[1]	mg/L	10	20
TSS ^[1]	mg/L	10	20

^[1] The average monthly percent removal of BOD₅ and TSS, as measured at Monitoring Location EFF-002, shall not be less than 85 percent.

The technology-based effluent limitations for BOD and TSS for the discharge to the Pajaro River at Discharge Point 002 are more stringent than required by the Secondary Treatment Regulations and reflect effluent quality achievable through tertiary treatment.

C. Water Quality-Based Effluent Limitations (WQBELs)

1. Scope and Authority

NPDES regulations at 40 CFR 122.44 (d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards, including numeric and narrative objectives within a standard.

The process for determining “reasonable potential” and calculating WQBELs, when necessary, is intended to protect the designated uses of receiving waters as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in the Basin Plan and in other applicable State and federal

rules, plans, and policies, including applicable water quality criteria from the CTR and the NTR.

Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBELs must be established in accordance with the requirements of 40 CFR 122.44 (d) (1) (vi), using (1) USEPA criteria guidance under CWA section 304 (a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed State criterion or policy interpreting the State's narrative criterion, supplemented with other relevant information.

2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

Beneficial uses described by the Basin Plan for the Pajaro River are presented in section II. H of the Order. Water quality criteria applicable to this receiving water are established by the CTR, the NTR, and by the Basin Plan. Because the discharge at Discharge Point 002 is the only discharge to surface waters, reasonable potential for pollutants with applicable water quality criteria was evaluated only for Discharge Point 002.

3. Determining the Need for WQBELs

NPDES regulations at 40 CFR 122.44 (d) require effluent limitations to control all pollutants which are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard.

The SIP, statewide policy that became effective on May 22, 2000, establishes procedures to implement water quality criteria from the NTR and CTR and for priority, toxic pollutant objectives established in the Basin Plan. The implementation procedures of the SIP include methods to determine reasonable potential (for pollutants to cause or contribute to excursions above State water quality standards) and to establish numeric effluent limitations, if necessary, for those pollutants which show reasonable potential.

The SIP Section 1.3 requires the Regional Board to use all available valid, relevant, and representative receiving water and effluent data and information to conduct a reasonable potential analysis. During the term of the previous permit, effluent data for the discharge at Discharge Point 002 was not available, because there were no discharges at Discharge Point 002 during the previous permit term. The Discharger has, however, collected annual monitoring data of its secondary effluent for discharge to land application at Discharge Point 001 for the Title 22 pollutants and additional quarterly monitoring data for some toxic pollutants. This data was used as representative effluent data in conducting the RPA for discharges to the Pajaro River at Discharge Point 002.

Some freshwater water quality criteria for metals are hardness dependent; i.e., as hardness decreases, the toxicity of certain metals increases and the applicable

water quality criteria become correspondingly more stringent. Regional Water Board staff used hardness data collected by the Central Coast Ambient Monitoring Program for the Pajaro River at Betabel Road, which is located immediately downstream from Discharge Point 002. A geometric mean of 323 mg/L was calculated using 15 hardness data points collected from January 2005 to March 2006 and used to determine hardness-based criteria.

To conduct the reasonable potential analysis, the Water Board identified the maximum observed effluent (MEC) and background (B) concentrations for each priority toxic pollutant from receiving water and effluent data provided by the Discharger and compared these data to the most stringent applicable water quality criterion (C) for each pollutant from the NTR, CTR, and the Basin Plan. Section 1.3 of the SIP establishes three triggers for a finding of reasonable potential.

Trigger 1. If the MEC is greater than C, there is reasonable potential, and an effluent limitation is required.

Trigger 2. If B is greater than C, and the pollutant is detected in effluent (MEC > ND), there is reasonable potential, and an effluent limitation is required.

Trigger 3. After reviewing other available and relevant information, a permit writer may decide that a WQBEL is required. Such additional information may include, but is not limited to: the facility type, the discharge type, solids loading analyses, lack of dilution, history of compliance problems, potential toxic impact of the discharge, fish tissue residue data, water quality and beneficial uses of the receiving water, CWA 303 (d) listing for the pollutant, and the presence of endangered or threatened species or their critical habitat.

Based on analysis of effluent data, the Regional Water Board, using methods presented in the SIP, finds that the discharge to the Pájaro River shows reasonable potential to cause or contribute to in-stream excursions above applicable water quality criteria for copper and chlorodibromomethane.

The following table summarizes the RPA for each priority toxic pollutant that was detected in effluent during monitoring events from 2004 through 2008. No other pollutants with applicable numeric water quality criteria from the NTR, CTR, and the Basin Plan (including the Title 22 pollutants) were measured above detectable concentrations during that monitoring period.

Table F-7. RPA Results

Pollutant	Units	C (Basis)	MEC	B	RPA Result
Copper	µg/L	25 (CTR freshwater chronic aquatic life)	50	Not Available	Yes
Nickel	µg/L	100 (Basin Plan [Title 22] human health)	4.5	Not Available	No
Chlorodibromomethane	µg/L	0.40 (CTR human health)	1.3	Not Available	Yes
Chloroform	µg/L	No Criteria	1.8	Not Available	No

Pollutant	Units	C (Basis)	MEC	B	RPA Result
Dichlorobromomethane	µg/L	0.56 (CTR human health)	0.53	Not Available	No
Aluminum	µg/L	1000 (Basin Plan [Title 22] human health)	70	Not Available	No
Barium	µg/L	1000 (Basin Plan [Title 22] human health)	66	Not Available	No
Fluoride	µg/L	1000 (Basin Plan Table 3-4 for agriculture)	600	Not Available	No
Nitrate as NO3	µg/L	45,000 (Basin Plan [Title 22] human health)	16,800	Not Available	No
Nitrate + Nitrite as N	µg/L	10,000 (Basin Plan [Title 22] human health)	3800	Not Available	No
Manganese	µg/L	200 (Basin Plan Table 3-4 for agriculture)	20	Not Available	No
Strontium-90	pCi/L	8 (Basin Plan [Title 22] human health)	1.15	Not Available	No
H-3	pCi/L	20,000 (Basin Plan [Title 22] human health)	313	Not Available	No
Ra-226/228	pCi/L	5 (Basin Plan [Title 22] human health)	1	Not Available	No
Gross Alpha	pCi/L	15 (Basin Plan [Title 22] human health)	2.96	Not Available	No
Gross Beta	pCi/L	No Concentration-based Criterion	29.1	Not Available	No
Uranium	pCi/L	20 (Basin Plan [Title 22] human health)	0.0823	Not Available	No

4. WQBEL Calculations

Final WQBELs for copper and chlorodibromomethane have been determined using the methods described in Section 1.4 of the SIP.

Step 1: For each water quality criterion/objective, an effluent concentration allowance (ECA) is calculated from the following equation to account for dilution and background levels of each pollutant.

$$ECA = C + D (C - B), \text{ where}$$

C = the applicable water quality criterion (adjusted for receiving water hardness and expressed as total recoverable metal, if necessary)

D = the dilution credit (here D = 0, as the Water Board has no information with which to justify credit for dilution)

B = the background concentration

Step 2: For each ECA based on an aquatic life criterion (i.e., copper), the long-term average discharge condition (LTA) is determined by multiplying the ECA

times a factor (multiplier), which adjusts the ECA to account for effluent variability. The multiplier varies depending on the coefficient of variation (CV) of the data set and whether it is an acute or chronic criterion/objective. Table 1 of the SIP provides pre-calculated values for the multipliers based on the value of the CV. When the data set contains less than 10 sample results, or 80 percent or more of the data are reported as non-detect (ND), the CV is set equal to 0.6. Derivation of the multipliers is presented in Section 1.4 of the SIP.

From Table 1 of the SIP, multipliers for calculating LTAs at the 99th percentile occurrence probability are 0.28 (acute multiplier) and 0.48 (chronic multiplier). LTAs are determined as follows.

Table F-8. Calculation of Long-Term Averages

Pollutant	ECA		ECA Multiplier		LTA (µg/L)	
	Acute	Chronic	Acute	Chronic	Acute	Chronic
Copper	42	25	0.28	0.48	11.8	12.2

Step 3: WQBELs, including an average monthly effluent limitation (AMEL) and a maximum daily effluent limitation (MDEL) are calculated using the most limiting (the lowest) LTA. The LTA is multiplied times a factor that accounts for averaging periods and exceedance frequencies of the effluent limitations, and for the AMEL, the effluent monitoring frequency. Here, the CV is calculated from the effluent data set as 0.71, and the sampling frequency is set equal to 4 ($n = 4$). The 99th percentile occurrence probability was used to determine the MDEL multiplier and a 95th percentile occurrence probability was used to determine the AMEL multiplier. From Table 2 of the SIP the MDEL multiplier is 3.58 and the AMEL multiplier is 1.66. Final WQBELs for copper are calculated as follows.

Table F-9. Calculation of Aquatic Life WQBELs

Pollutant	LTA	MDEL Multiplier	AMEL Multiplier	MDEL (µg/L)	AMEL (µg/L)
Copper	11.8	3.58	1.66	42	20

Step 4: When the most stringent water quality criterion is a human health criterion (i.e., chlorodibromomethane), the AMEL is set equal to the ECA, and the MDEL is calculated by multiplying the ECA times the ratio of the MDEL multiplier to the AMEL multiplier.

From Table 2 of the SIP, when $CV = 0.6$ and $n = 4$, the MDEL multiplier at the 99th percentile occurrence probability equals 3.11, and the AMEL multiplier at the 95th percentile occurrence probability equals 1.55. Final WQBELs for chlorodibromomethane are presented below.

Table F-10. Calculation of Human Health WQBELs

Pollutant	ECA	MDEL/AMEL Multiplier	MDEL (µg/L)	AMEL (µg/L)
Chlorodibromomethane	0.40	$3.11/1.55 = 2.01$	0.80	0.40

5. Whole Effluent Toxicity (WET)

Whole effluent toxicity (WET) limitations protect receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. WET tests measure the degree of response of exposed aquatic test organisms to an effluent. The WET approach allows to protect the narrative “no toxics in toxic amounts” criterion while implementing numeric criteria for toxicity. There are two types of WET tests - acute and chronic. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth.

The Basin Plan specifies a narrative objective for toxicity, requiring that all waters be maintained free of toxic substances in concentrations that are toxic to, or which produce detrimental physiological responses in human, plant, animal, or aquatic life. Survival of aquatic organisms in surface waters subjected to a waste discharge or other controllable water quality conditions shall not be less than that for the same water body in areas unaffected by the waste discharge or for another control water.

The previous permit did not include numeric effluent limitations for whole effluent toxicity, but it required monitoring for whole effluent acute and chronic toxicity, at Discharge Point 002, including accelerated monitoring when acute toxicity or chronic toxicity was measured above one toxicity unit. Because there was not a discharge at Discharge Point 002 during the term of the previous permit, whole effluent toxicity monitoring was not conducted. Due to the variable character of municipal wastewater, however, the Regional Water Board has determined that monitoring should continue. This Order therefore retains the effluent monitoring requirements at Discharge Point 002, and requires accelerated monitoring when acute toxicity is detected in the effluent or when the chronic toxicity trigger value of one TUC is exceeded.

6. **TDS, Sodium, Chloride, Sulfate, and Boron.** Final effluent limitations for TDS, sodium, chloride, sulfate, and boron at Discharge Point 002 are retained from the previous permit, and reflect applicable water quality criteria established by Table 3-7 of the Basin Plan for the Pájaro River at Chittenden.
7. **Nitrate Nitrogen.** This permit retains the effluent limitations for nitrate nitrogen at Discharge Point 002. The limitations are based on Title 22 MCLs.
8. **Unionized Ammonia.** Effluent limitations for unionized ammonia at Discharge Point 002 are retained from the previous permit and reflect the WQO for ammonia established by section II.A.2 of the Basin Plan for all inland surface waters of the Region, applied as end-of-pipe effluent limitations.
9. **Chlorine, Total Coliform Bacteria, and Turbidity.** WQBELs for chlorine, total coliform bacteria, and turbidity at Discharge Point 002 are retained from the previous permit, and reflect Title 22 recycled water requirements for disinfected secondary-2.2 recycled water production.

10. pH. Effluent limitations for pH are established by the Order for discharges at Discharge Point 002, and are based in the Basin Plan to protect receiving water beneficial uses.

11. Additional Discharge Specifications. Additional discharge specifications applicable to Discharge Point 002 are retained from the previous permit.

A summary of all WQBELs applicable at Discharge Point 002 are presented in Table F-11 below.

Table F-11. Summary of WQBELs – Discharge Point 002

Constituent	Units	Effluent Limits		
		AMEL	MDEL	12-Month Moving Avg
Nitrate as N	mg/L	5	10	---
Unionized Ammonia as N	mg/L	0.025	0.050	---
pH	s.u.	6.5 – 8.3 at all times		---
TDS	mg/L	1000	---	---
Sodium	mg/L	200	---	---
Chloride	mg/L	250	---	---
Sulfate	mg/L	250	---	---
Copper	µg/L	20	42	---
Lead	µg/L	2.1	4.2	---
Chlorodibromomethane	µg/L	0.40	0.80	---
Turbidity	NTU	Daily turbidity shall be less than or equal to 2 NTU. Turbidity shall not exceed 10 NTU at any time and shall not exceed 5 NTU for more than 5 percent of the time.		
Chlorine	mg/L	Total residual chlorine shall be undetectable at any time as determined by amperometric titration or another equally sensitive method. A CT value of not less than 45 mg-min/L shall be maintained at all times.		
Total Coliform Bacteria	MPN/100 mL	The median most probable number (MPN) of total coliform organisms in effluent shall not exceed 2.2 MPN/100 mL, based on the results of the last seven days for which samples have been collected. The MPN of total coliform organisms shall not exceed 23 MPN/100 mL in more than one sample in a 30-day period, and not single sample shall exceed 240 MPN/100 mL at any time.		

D. Final Effluent Limitations

Final, technology-based and water quality-based effluent limitations established by the Order are discussed in the preceding sections of the Fact Sheet.

1. Satisfaction of Anti-Backsliding Requirements

The Order satisfies applicable anti-backsliding provisions of the Clean Water Act, as all limitations and requirements of the Order are at least as stringent as those of the previous permit.

Effluent limitations for thallium, chloroform, bromodichloromethane, bis(2-ethylhexyl)phthalate, aluminum, and manganese are not retained, as these pollutants no longer demonstrate reasonable potential. Elimination of effluent limitations that do not show reasonable potential is consistent with the anti-backsliding exception at CWA 402(o) (2) (B) (i), where information is available which was not available at the time of permit issuance. Here, the effluent data generated during the term of the previous permit indicate that these pollutants are no longer discharged at concentrations that may cause or contribute to exceedances of applicable water quality criteria.

Effluent data were not available for lead and therefore the effluent limitations for this pollutant are retained from the previous permit, until monitoring data become available for future RPAs.

2. Satisfaction of Antidegradation Policy

Provisions of the Order are consistent with applicable anti-degradation policy expressed by NPDES regulations at 40 CFR 131.12 and by State Water Board Resolution No. 68-16. Limitations and conditions of the Order assure maintenance of the existing quality of receiving waters and do not authorize increased rates of discharge or increased pollutant loadings to the receiving water above that authorized by the previous Order.

3. Stringency of Requirements for Individual Pollutants

This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based effluent limitations consist of restrictions on TSS and BOD₅. Restrictions on these pollutants are discussed in Section IV.B of the Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum federal technology-based requirements that are necessary to meet water quality standards. These limitations are not more stringent than required by the CWA.

Final, technology and water quality based effluent limitations are summarized in sections IV.B and C of this Fact Sheet.

E. Interim Effluent Limitations

The Order does not establish interim effluent limitations and schedules for compliance with final effluent limitations.

F. Land Discharge Specifications

1. BOD and TSS.

Numeric limitations for BOD and TSS, including an 85 percent removal requirement, for the land application at Discharge Point 001 are retained from the previous permit and reflect wastewater quality achievable by secondary treatment.

2. TDS, Sodium, Chloride, Sulfate, and Boron.

Final effluent limitations for TDS, sodium, chloride, sulfate, and boron at Discharge Point 001 are retained from the previous permit and are based on interpretation of Basin Plan Tables 3-3 and 3-4 for the protection of agricultural irrigation uses of groundwater.

3. Nitrate Nitrogen.

Effluent limitations for nitrate nitrogen at Discharge Point 001 are retained from the previous permit and are based on the Title 22 MCL for nitrate as nitrogen.

4. pH.

Effluent limitations for pH at Discharge Point 001 are retained from the previous Order.

5. Additional Discharge Specifications.

Additional operational and engineering specifications for Discharge Point 001 are retained from the previous permit.

A summary of land discharge numeric specifications applicable to Discharge Point 001 are summarized in the table below.

Table 11. Summary of Land Discharge Specifications.

Constituent	Units	Effluent Limits			
		AMEL	AWEL	MDEL	12-Month Moving Avg
BOD ₅ ^[1]	mg/L	30	45	---	---
TSS ^[1]	mg/L	30	45	---	---
Nitrate as N	mg/L	5	---	10	---
pH	s.u	6.5 – 8.3 at all times			---
TDS	mg/L	---	---	---	900
Chloride	mg/L	---	---	---	200
Sodium	mg/L	---	---	---	175
Sulfate	mg/L	---	---	---	150
Boron	mg/L	---	---	---	1.0

[1] The average monthly removal of BOD₅ and TSS shall not be less than 85 percent.

G. Reclamation Specifications

Reclamation use of tertiary treated wastewater shall adhere to applicable requirements of CWC sections 13500 – 13577 (Water Reclamation) and of CCR Title 22, sections 60301 – 60357 (Water Recycling Criteria). The Discharger's reclamation system is regulated under Regional Water Board Master Water Reclamation Requirements Order No. 98-052.

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

A. Surface Water

Receiving water quality is a result of many factors, some unrelated to the discharge. This Order considers these factors and is designed to minimize the influence of the discharge on the receiving water. Specific water quality objectives established by the Basin Plan to meet this goal for all inland surface waters are included as Receiving Water Limitations in Section V.A of this Order.

B. Groundwater

Groundwater limitations established by the Order include general objectives for groundwater established by the Basin Plan.

VI. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

NPDES regulations at 40 CFR 122.48 require that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 also authorize the Regional Water Board to require technical and monitoring reports. Rationale for the monitoring and reporting requirements contained in the Monitoring and Reporting Program (MRP), which is presented in Attachment E of this Order, is presented below.

A. Influent Monitoring

The influent monitoring requirements are unchanged and are retained from the previous Order.

B. Effluent Monitoring

At EFF-001, all monitoring requirements are unchanged and are retained from the previous Order.

At EFF-002, most effluent monitoring requirements are unchanged and are retained from the previous Order, except for the following.

- Routine monthly monitoring for thallium, chloroform, dichlorobromomethane, and bis(2-ethylhexyl)phthalate is not retained because these pollutants are no longer required as demonstrated by the reasonable potential analysis. It should be noted that monitoring of these parameters is required once per permit term as part of the CTR pollutant scan.

- Monitoring requirements for fecal coliform bacteria were established in the MRP for determination of compliance with the discharge prohibition at Discharge Point 002 against any discharge of fecal coliform bacteria originating from human sources.

C. Whole Effluent Toxicity Testing Requirements

Whole effluent toxicity (WET) limitations protect receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. Acute toxicity testing measures mortality in 100 percent effluent over a short test period and chronic toxicity testing is conducted over a longer period of time and may measure mortality, reproduction, and or growth. This Order retains acute and chronic toxicity monitoring requirements for Discharge Point 002 from the previous permit.

D. Receiving Water Monitoring

1. Surface Water

Most receiving water and surface water monitoring requirements are unchanged and are retained from the previous Order. The MRP establishes monitoring requirements for the Title 22 pollutants to generate background data for future RPA for these pollutants.

2. Groundwater

Groundwater monitoring requirements are unchanged and are retained from the previous Order.

E. Other Monitoring Requirements

1. Biosolids monitoring requirements are unchanged and are based on 40 CFR Part 503.
2. Pond maintenance requirements are retained from the previous Order.

VII. RATIONALE FOR PROVISIONS

A. Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR 122.42, are provided in Attachment D to the Order.

NPDES regulations at 40 CFR 122.41 (a) (1) and (b - n) establish conditions that apply to all state-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. 40 CFR 123.25 (a) (12) allows the State to omit or modify conditions to impose more stringent requirements. In accordance with 40 CFR 123.25, this Order omits federal conditions that address enforcement authority specified in 40 CFR 122.41 (j) (5) and (k) (2), because the enforcement authority under the

Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387 (e).

B. Monitoring and Reporting Program (MRP) Requirements

The Discharger shall comply with the MRP, and future revisions thereto, in Attachment E of this Order.

C. Special Provisions

1. Reopener Provisions

The Order may be modified in accordance with the requirements set forth at 40 CFR 122 and 124, to include appropriate conditions or limits based on newly available information, or to implement any, new State water quality objectives that are approved by the USEPA. As effluent is further characterized through additional monitoring, and if a need for additional effluent limitations becomes apparent after additional effluent characterization, the Order will be reopened to incorporate such limitations.

2. Special Studies and Additional Monitoring Requirements

The Order includes the requirement to conduct accelerated whole effluent toxicity monitoring upon the detection of acute toxicity in the effluent, or upon the exceedance of the chronic toxicity effluent limitation.

3. Best Management Practices and Pollution Prevention

The Order does not establish requirements regarding best management practices and pollution prevention.

4. Construction, Operation, and Maintenance Specifications

The Order does not establish construction, operation, or maintenance specifications.

5. Special Provisions for Municipal Facilities (POTWs Only)

a. Biosolids Management

Provisions regarding sludge handling and disposal ensure that such activities will comply with all applicable regulations.

40 CFR Part 503 sets forth USEPA's final rule for the use and disposal of biosolids, or sewage sludge, and governs the final use or disposal of biosolids. The intent of this federal program is to ensure that sewage sludge is used or disposed of in a way that protects both human health and the environment.

USEPA's regulations require that producers of sewage sludge meet certain reporting, handling, and disposal requirements. As the USEPA has not delegated the authority to implement the sludge program to the State of

California, the enforcement of sludge requirements that apply to the Discharger remains under USEPA's jurisdiction at this time. USEPA, not the Water Board, will oversee compliance with 40 CFR Part 503.

b. Pretreatment

Pretreatment requirements for POTWs are contained within 40 CFR Part 403. Per 40 CFR Part 403.8, any POTW (or combination of POTWs operated by the same authority) with a total design flow greater than 5 million gallons per day (MGD) and receiving from industrial users pollutants which pass through or interfere with the operation of the POTW or are otherwise subject to pretreatment standards will be required to establish a POTW pretreatment program unless the NPDES State exercises its option to assume local responsibilities as provided for in §403.10(e). The Executive Officer may require that a POTW with a design flow of 5 MGD or less develop a POTW pretreatment program if he or she finds that the nature or volume of the industrial influent, treatment process upsets, violations of POTW effluent limitations, contamination of municipal sludge, or other circumstances warrant in order to prevent interference with the POTW or pass through as defined in 40 CFR Part 403.3.

The Order retains pretreatment requirements as the Facility has total effluent flows in excess of 5 MGD and a number of significant industrial users.

c. Infiltration/Inflow and Spill Prevention Program Requirements

Infiltration/inflow and spill prevention program requirements are not retained from the previous permit as the Discharger is required to enroll in the Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, State Water Board Order No. 2006-0003-DWQ.

6. Other Special Provisions

a. Storm Water

The Order does not address discharges of storm water from the treatment and disposal site, except to require coverage by and compliance with applicable provisions of General Permit No. CAS000001 - *Waste Discharge Requirements for Discharges of Storm Water Associated with Industrial Activities Excluding Construction Activities*, if applicable.

b. Statewide General Waste Discharge Requirements for Sanitary Sewer Systems (State Water Board Order No. 2006-0003-DWQ)

This General Permit, adopted on May 2, 2006, is applicable to all "federal and state agencies, municipalities, counties, districts, and other public entities that own or operate sanitary sewer systems greater than one mile in length that collect and/or convey untreated or partially treated wastewater to a publicly owned treatment facility in the State of California." The purpose of the General Permit is to promote the proper and efficient management, operation, and

maintenance of sanitary sewer systems and to minimize the occurrences and impacts of sanitary sewer overflows. The Order requires the Discharger to seek coverage under the General Permit, if applicable, and comply with its requirements.

7. Compliance Schedules

The Order does not establish interim effluent limitations and schedules for compliance with final effluent limitations.

VIII. PUBLIC PARTICIPATION

The Regional Water Board (Water Board) is considering the issuance of waste discharge requirements (WDRs) that will serve as a National Pollutant Discharge Elimination System (NPDES) permit for the South County Regional Wastewater Authority Wastewater Treatment Facility. As a step in the WDR adoption process, the Water Board staff has developed tentative WDRs. The Water Board encourages public participation in the WDR adoption process.

A. Notification of Interested Parties

The Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations.

In a December 7, 2009 letter, Regional Water Board staff informed the discharge of our intent to have the March 18, 2010 meeting. The letter also transmitted instructions (and a Public Notice) for the discharge to publish in a local newspaper. The discharger published the Public Notice on December 22, 2009 in the Morgan Hill Times stating that comments were due by January 21, 2010.

B. Written Comments

SCRWA's comments and staff's responses follow:

1. SCRWA requests that re-numbering of the monitoring wells be avoided if possible. The new order revises the numbers for all the monitoring wells, re-tabulates them, and differentiates water elevation vs. water quality monitoring using footnotes rather than the separate tables in the previous permit. This re-numbering will require work by the treatment plant staff to add new marking to the wellheads and revise field procedures and record-keeping databases. It will complicate comparisons of past and future data summaries and reports. The existing numbers on well drilling logs and construction diagrams cannot be changed retroactively. Based on past experience, it will be necessary to keep track of both the old and new numbers to acquaint new personnel with the numbering change. For example, MW12/PWV will become GW-009/MW12/PWV. The listing sequence used on existing permit tables generally groups the wells chronologically by installation date, which correlates with well construction characteristics and associated percolation pond operational groups. The new sequence does not appear to follow any set pattern. The new

sequence re-uses well numbers (e.g. well number 23) that are also used for existing wells that are not included in the monitoring program. Note that new well number GW-017 is assigned to two different wells (PWI and PWL) while new number GW-018 is not used. It is suggested that the intent was to renumber PWL as GW-018. However, SCRWA recommends that the need to resolve this kind of error be reduced by avoiding the renumbering entirely.

Staff Response: The proposed Order does not renumber the groundwater monitoring wells. The proposed Order assigns new monitoring location numbers to each well but retains the monitoring well identification numbers. Table E-1 provides both numbers for each well. The Discharger agreed to include both numbers in self-monitoring reports to the Executive Officer.

2. There are two redundant sections relating to salt management. One is repeated from the previous permit and just addresses salts. The second appears to be a more standardized version that addresses nutrients (specifically nitrogen) as well as salts. Assuming that submittal of two salt management reports is not necessary, these permit sections should be edited to remove this redundant requirement. The due dates for the reports should also be reconsidered. While submittal of an annual data summary report by January 30 may be practical, the March due date is more appropriate for analysis and development of recommended actions. Looking beyond these details, SCRWA concurs with the discussion in Section II – S “Recycled Water Policy” on page 9 of the order which explains that a basin-wide strategic planning approach is intended instead of “imposing requirements on individual projects”. Considering that the existing SCRWA plant is already a nitrification/denitrification facility, it is questionable why the new SCRWA permit should require an annual evaluation of the need for installing nitrification and denitrification. It appears that the actual objective of the requirement is to initiate a coordinated salt and nutrient management effort by appropriate local agencies including, for example, the Santa Clara Valley Water District (SCVWD). SCRWA requests confirmation and clarification of this objective (e.g. by explanation in Attachment F Fact Sheet) along with removal of the salt management reporting redundancies.

Staff Response. The proposed Order's Section C.8 requires the Discharger to either develop and implement a salt and nutrient reduction program or to participate in a regional program. If the Discharger were to develop its own program, the proposed Order required the following task be performed:

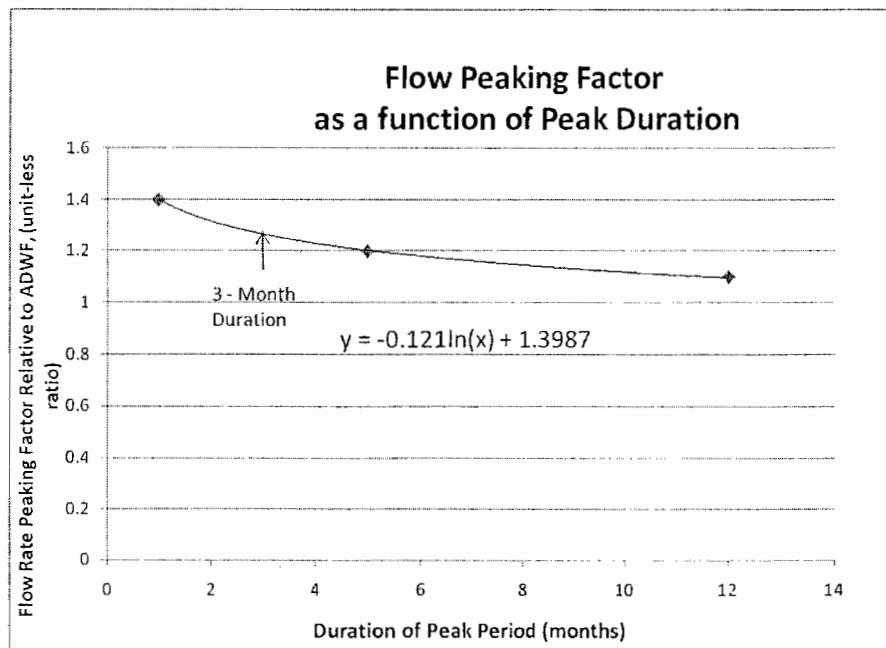
“Analysis of wastewater treatment facility ability to facilitate nitrification and denitrification, or other means of nitrogen removal”. The Discharger's treatment plant now nitrifies and denitrifies the wastewater so the requirement is unneeded and staff removed it from the Provision.

However, note that the Discharger will likely participate with the Santa Clara Valley Water District (District) in a regional salt and nutrient management program, as described in the District's comments, discussed below.

After determining that the requirements in the salt and nutrient management plan

described in the proposed Order's Section C.8 addressed the requirements of the salt management plan in the existing Order, staff removed the proposed Order's Provision C.6, which required the Discharger implement a salts management program.

3. Discharge prohibition III.F defines the Average Dry Weather Flow (ADWF) based on the three driest months in a year, and then limits the average flow for the three wettest months to 10.2 mgd, which is equal to 1.2 times the ADWF. This 1.2 factor was originally calculated based on the averages of the 5 wettest months, as documented in reports previously submitted to the RWQCB (e.g. the Effluent Management Plan (2004) and the annual Hydrologic Balance reports submitted prior to 2005). These reports showed that the SCRWA facility can handle the projected wet season hydrograph corresponding to the 5-month factor. The change in definition to a 3-month factor arbitrarily lowers the limiting hydrograph curve, increasing the probability that flows will exceed the wet weather flow limit before the plant reaches its actual capacity. As documented in the Capacity Report submitted to support re-rating the plant in 2007, the treatment plant is capable of treating a sustained peak monthly flow of 11.9 mgd (corresponding to a 1.4 peaking factor). The attached graph calculates a 3-month wet season peaking factor value of 1.27 by interpolation between the 1-month and 5-month factors. Using this factor, the appropriate flow limit for the three wettest months can be calculated to be 10.8 mgd. SCRWA requests that this value be used in the permit.



Staff response: The Discharger reconsidered the engineering basis for the influent flow limitation specified in the proposed Order at Discharge Prohibition III. F. Staff concurs that the influent flow limitation based on the reconsideration should be 10.8 mgd and changed the prohibition accordingly, from 10.2 mgd.

4. New effluent limits for copper (20 µg/L monthly, 42 µg/L daily per Table 7 on page 12 of the order) are proposed for discharge location 002 (Pájaro River). These limits

are generated by the RPA calculations even though the secondary effluent copper was reported as "below detection" in the application, because the reporting limit is higher than the applicable water quality standard. Secondary effluent data collected in March, 2009 (after the application data period) showed copper at 4.6 µg/L, which is below the MEC of 50. These values are substantially below the proposed limits, so compliance should not be an issue in the tertiary treated discharge. Copper has not been regulated under Title 22 in the recycled water permit, so data is not available for tertiary effluent. While it could be argued that there is little justification for the copper limit, copper is an important and common wastewater constituent and it is probably in the best interest of SCRWA that analytical data be collected for it in the event of a discharge. SCRWA thus concurs with addition of this limit.

Staff Response. Comment noted.

The Santa Clara Valley Water District's comments and staff's response follows:

The District stated that it shall develop a salt and nutrient management plan for all of Santa Clara County in accordance with the State Recycling Policy and they plan to include SCRWA in the plan. SCWRA staff stated that they agree with this approach. Regional Board staff will participate in plan development and ensure its timely progress.

C. Public Hearing

The Regional Water Board held a public hearing on the tentative WDRs during its regular meeting on the following date and time and at the following location:

Date: **March 18, 2010**
Time: **8:30 a.m.**
Location: **Watsonville City Council Chambers**
275 Main Street, 4th Floor
Watsonville, CA 95076

D. Waste Discharge Requirements Petitions

Any aggrieved person may petition the State Water Resources Control Board to review the decision of the Water Board regarding the final WDRs. The rules regarding petitions are found in the California Code of Regulations, Title 23, Sections 2050-2068. The petition must be submitted within 30 days of the Water Board's action to the following address:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

E. Information and Copying

The Report of Waste Discharge (ROWD), related documents, tentative effluent limitations and special provisions, comments received, and other information are on file and may be

inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Water Board by calling **805-549-3147**.

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this facility, and provide a name, address, and phone number.

G. Additional Information

Requests for additional information or questions regarding this order should be directed to **Michael Higgins** at **(805)542-4649** or **MHiggins@waterboards.ca.gov**.

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